

Applying Explainable AI Techniques on ML & DL Models in the Chest X-ray Domain

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Abstract. This study explores the application of Explainable Artificial Intelligence (XAI) in enhancing the interpretability of Machine Learning (ML) and Deep Learning (DL) models for multi-label classification of thoracic diseases using chest X-ray (CXR) images. Utilising the NIH Chestx-ray14 dataset, which includes over 112,000 annotated CXR images across 14 disease categories, we implemented and evaluated a range of models, including Kernel SVM, DenseNet121, MobileNetV2, SwinCheX, AE-CNN, and MLRFNet. Performance was evaluated using Area Under the Curve (AUC) metrics, with model scores ranging from 0.519 to 0.9987. To support clinical adoption and model transparency, XAI techniques Grad-CAM, LIME, SHAP, Eigen-CAM, and saliency maps, were applied to interpret model predictions and highlight relevant regions. The study focuses on the practical significance of integrating XAI in medical imaging pipelines to improve diagnostic decision-making.

Index Terms: Chest X-ray, Thoracic Disease Classification, Explainable AI, Machine Learning, Deep Learning, Grad-CAM, LIME, Multi-label Classification

I. Introduction

Thoracic disease classification from chest X-ray (CXR) images is vital for early diagnosis, particularly for lung cancer, which accounts for 11-12% of cancer cases and 18% of cancer-related deaths [1]. AI models, including Convolutional Neural Networks (CNNs) and transfer learning, offer high accuracy but lack interpretability, hindering clinical trust. Nonetheless, when deploying AI models for real-world use, models must provide adequate reasons for their judgments under the European Union's General Data Protection Regulation (GDPR) rule [2]. This legislation highlights the value of transparent decision-making processes. Thus, ensuring transparent clinical judgements using AI models necessitates a thorough grasp of the model's fundamental workings [3]. This study applies XAI techniques to ML and DL models using

the NIH Chestx-Ray14 dataset to classify 14 thorax diseases, aiming to balance performance and transparency for radiological assessments.

The rest of our document is structured as follows. Section II discusses similar works in the thoracic disease classification tasks. Section III describes our methodology in full. In Section IV, this section includes an overview of the experimental results. Section V analyses the study's objectives and accomplishments, stressing the challenges of the existing work and identifying the effectiveness of our interpretability techniques. In section VI, we conclude this project.

II. Related Work

In this section, we present a comprehensive review of existing research in the field of thoracic disease classification using the ChestX-Ray14 dataset [4]. The review is organized into sub-sections as follows. We begin with traditional Machine Learning approaches (SVM), followed by CNN and Transfer Learning techniques. Furthermore, we explore recent developments in Transformers and Autoencoder Frameworks. Then, we highlight methods integrating Attention Mechanisms to enhance classification performance. Finally, we discuss the growing importance of Explainability and Interpretability in medical imaging models, particularly in high-stakes diagnostic environments.

Machine learning

In recent studies, the application of machine learning for thoracic disease classification using chest X-ray images has gained significant attention. As [5] proposed, a Kernel Support Vector (SVM)-based framework combined with Fast Principal Component Analysis (FPCA) was used to classify 14 thoracic diseases using the NIH ChestX-ray dataset, comprising 112,120 frontal-view images. FPCA was used to enhance computational efficiency compared to traditional PCA. In [5], four SVM kernels were evaluated: Linear (L-SVM), Polynomial (P-SVM), Radial Basis Function (RBF-SVM), and Hyperbolic Tangent (H-SVM). Results show that P-SVM and RBF-SVM achieved the highest performance, with validation ranging from 92% to 98% and accuracy nearing 100%. Monte Carlo cross-validation and ANOVA confirmed that Polynomial and RBF kernels are optimal for disease detection. The limitations were handling multilabel classification and computational demands for large datasets.

CNN and Transfer Learning

Previous studies have explored different deep learning approaches for thorax disease classification using CXR images. CNN has significantly improved accuracy by automating feature extraction. Several methods utilized pre-trained models like ResNet, VGG-Net, and DenseNet, with CheXNet (a fine-tuned DenseNet-121) achieving state-of-the-art performance. Researchers also investigated multi-label classification techniques, such as two-stage neural networks and attention-based models, to capture label dependencies. However, these methods faced challenges like high computational power, long training time, and label imbalance issues. This paper explores problem transformation methods like binary relevance, label powerset, and classifier chains, which offer a balance between computational efficiency and performance [6].

Deep learning has improved the automatic classification of lung diseases using chest X-rays, with convolutional neural networks (CNN) playing an important role. Several studies have explored different architectures, such as DenseNet, VGG19, and hybrid models, to enhance classification performance. The ChestX-ray14 dataset, which has been used for developing diagnostic models. With CNN with recurrent networks, and transfer learning, classification has improved, and

by optimizing techniques and attention mechanisms, to achieve high accuracy in detecting lung diseases. The MobileNetV2 architecture, known for its efficiency in low-power devices, has proved its reliability in medical imaging applications, making it a reliable model for lung disease classification [7]. Previous studies have used deep learning models like DenseNet, ResNet, and MobileNetV2 for lung disease classification from chest X-rays, achieving promising but limited results due to data imbalance and small sample sizes. To overcome these issues, this study fine-tunes MobileNetV2 into MobileLungNetV2, enhancing image quality and using data augmentation to improve accuracy, outperforming previous models in multiclass lung disease classification [8].

In [9], they introduced NSGANetV1, a multi-objective NAS framework that optimizes Convolutional Neural Network (CNN) for both classification accuracy and floating-point operations (FLOPs). The method introduces a hybrid strategy combining genetic algorithms (exploration) and Bayesian network-based distribution estimation (exploitation) to approximate the Pareto frontier of architectures in a single run. Evaluated on benchmark datasets (CIFAR, ImageNet) and medical imaging tasks (chest X-ray classification), NSGANetV1 generated architectures that outperformed manually designed networks (e.g., ResNet, DenseNet) and automated approaches (e.g., RL-based NASNet, AmoebaNet). Notably, NSGANetV1 achieved state-of-the-art results on the ChestX-Ray14 dataset with 84.6% AUROC using $3.2\times$ fewer parameters than the prior best model. The framework demonstrated computational efficiency, requiring only 27 GPU-days for search, $10\text{--}100\times$ faster than existing NAS methods.

A custom convolutional neural network was developed to classify chest X-rays into three groups: Normal, Pneumonia, and Tuberculosis [10]. The main goal was to make diagnoses faster and more accessible, especially in places without radiologists. Their model, trained on a balanced dataset and preprocessed grayscale images, achieved an accuracy of 90 percent, precision, and recall. It gives results in just a few seconds. This speed and reliability could be useful in hospitals or clinics with high patient loads. The research shows how deep learning can be applied in real medical settings to save time and improve early detection.

Recent advancements in deep learning have significantly enhanced thoracic disease classification using chest X-ray images. In [11], they pioneered the use of pre-trained image classifiers on non-medical datasets for chest pathology detection, demonstrating feasibility but limited to single-label classification. In [12], they achieved an AUC of 0.964 using GoogleNet on a medium-sized dataset of 35,000 X-rays, though their focus remained on single pathologies. In [13], they introduced CheXNet, a 121-layer DenseNet trained on 100,000 images, which surpassed radiologist performance in pneumonia detection (AUC = 0.896). In [4], they developed the ChestX-ray8 dataset, enabling weakly supervised multi-label classification of 14 pathologies, though their model struggled with co-occurring disease detection. Comparative studies (Tables 4-5) highlight DenseNet-based architectures outperforming earlier single-label models, with the proposed method achieving state-of-the-art AUC scores (e.g., 0.90 for Cardiomegaly). Key limitations include reliance on single-label frameworks, computational demands, and inadequate handling of class imbalance. While [17], [18] demonstrated high AUCs (0.99) for tuberculosis detection using ensemble CNNs, their datasets were small, limiting generalizability. These studies underscore the need for more advanced multi-label frameworks to address real-world clinical scenarios where patients often present with multiple pathologies.

Transformers

While CNNs have been the norm for image classification, Vision Transformers (ViTs) have higher accuracies than their traditional CNN counterparts. [19] The SwinCheX framework elaborates on this by using the Swin Transformer [20] as the backbone of its model, having Grad-Cam [21] for interpretability, and introducing windowed self-attention, which reduces computation by focusing on the local ViT instead of the global ViT. SwinCheX achieved a total of 0.81 AUC score compared to the 0.807 from the DNet [22].

Auto Encoders Frameworks

A model was developed to simultaneously compress and classify chest X-ray images [23]. Instead of resizing images and risking the loss of important details, they used an autoencoder to shrink the image while keeping useful features. That way, the classifier still has enough information to detect diseases. They tested this on a large dataset, ChestXray14, and it ended up doing better than some other models. It's smart to avoid handling large images without overloading memory or missing important features. The whole thing is built to be more efficient and accurate at the same time.

Although using basic CNNs like CheXNet to attention mechanisms like A³Net [24] and graph-based methods like CheXGAT [25] have gotten relatively high AUC scores. However, these models failed to capture the data in images that appeared in different image resolutions while not focusing on the class imbalance, which is why MLRFNet was created. MLRFNet is a multi-layer residual feature classifier used to classify different features of the image and maintain their details through 3 different resolutions, combining this with efficient channel attention (ECA) [26]. The MLRFNet achieved an 0.853 AUC on the ChestX-Ray14 dataset, maintaining a 46 ms/image speed, while the next highest models were CheXGat with 0.827 and A³Net with 0.826.

Attention Mechanism

A method was introduced to address the challenge of classifying chest X-rays that show multiple diseases in one image [27]. The approach named CRAL assigns separate attention to each disease rather than treating all labels equally. This helps the model keep an eye on features that are relevant to each condition and ignore unrelated ones. CNN models, such as ResNet and DenseNet, were used for feature extraction, with weights that were applied for each of the classes. The model was tested on the ChestXray14 dataset and showed a strong performance, especially in complex cases where multiple conditions were present in one scan.

Explainability and Interpretability

In [28], the paper presents a comprehensive survey of explainable artificial intelligence (XAI) methods for deep learning-based medical image analysis. The authors proposed a framework to categorize XAI techniques into model-based versus post hoc, model-specific versus model-agnostic, and global versus local explanations. This framework clarifies how different techniques- ranging from visual methods like saliency mapping (including CAM, Grad-CAM, and LRP) to textual and example-based explanations- can explain the "Black Box" nature of deep neural networks in medical imaging. The paper not only synthesizes current research trends in XAI for medical image analysis but also identifies future research directions aimed at improving the reliability and user trust in deep learning applications in high-stakes medical environments.

As stated, Deep Learning models for the X-ray dataset [2] have been going through constant improvement. However, they have not yet been implemented in a professional setting as they have not been interpretable enough. Vision-based models like CLIP Dissect[29], FALCON[30], and WWW[31], which all use Contrastive Language-Image Pretraining (CLIP), combine both Natural language processing for image labels and computer vision for images to give each image its text descriptors and their vector embeddings. [3] While having a CLIP-based similarity algorithm, it got a higher F1 score of 0.2619 (± 0.113) using its adaptive neuron selection in its Mask-free Medical Model Interpretation (MAMMI) compared to the FALCON with 0.0635 (± 0.027) and CLIP Dissect with 0.0714 (± 0.069).

III. Methodology

Tools and Software

In this research paper, we used many tools, including NumPy and Pandas for data handling, Matplotlib and Seaborn for visualisation, OpenCV, PIL, and skimage.io for image processing, and scikit-learn for preprocessing and evaluation metrics. Deep learning models were implemented using TensorFlow, PyTorch, and the Hugging Face Transformers library. Dataset manipulation was done using the datasets library. All experiments were done using Python 3.10 in Jupyter Notebook environments with GPU acceleration.

Dataset

[4] The dataset used in this study comprises 112,104 chest X-ray images, each accompanied by a comprehensive set of metadata stored in a data frame with 12 columns. These columns include: Image Index (unique identifier, object), Finding Labels (diagnoses, object), Follow-up # (integer), Patient ID (integer), Patient Age (integer), Patient Gender (object), View Position (object), OriginalImage[Width] (integer), Height (integer), OriginalImagePixelSpacing[x] (float), and y] (float). All images are stored in PNG format, ensuring high quality but resulting in larger file sizes than JPEG, a factor considered in the model pipeline development.

Images are resized to a standardised resolution of 1024×1024 pixels for consistency in model training, although original dimensions vary across the dataset and are preserved in the data frame. The Image Index follows the naming convention PatientID_ImageNumber.png (e.g., 00000001_000.png), where PatientID is the unique patient identifier, and ImageNumber represents the sequential image count for that patient. The highest follow-up number observed is 183, indicating that at least one patient has up to 183 images, reflecting extensive longitudinal data for certain individuals.

The Patient Age column raised concerns, as some entries suggest ages as high as 414. This anomaly is likely due to data entry errors or a non-standard scale, necessitating further validation to confirm whether ages are recorded in years or require correction. This issue will be addressed in preprocessing to ensure data integrity.

The Finding Labels column serves as the ground truth for model training, encompassing 15 classes: 14 disease categories (Atelectasis, Consolidation, Infiltration, Pneumothorax, Edema, Emphysema, Fibrosis, Effusion, Pneumonia, Pleural Thickening, Cardiomegaly, Nodule, Mass, Hernia) and a "No Findings" category. Notably, images may be associated with multiple labels, reflecting co-occurring conditions. For instance, labels such as Pneumonia|Effusion and Effusion|Pneumonia are equivalent, and accounting for such symmetries reduces the number of unique label combinations from 836 to 801, a reduction of approximately 30.

Analysis revealed that approximately 1,600 images are associated with four or more labels, which may indicate complex cases or potential labelling errors. These instances will be carefully reviewed during preprocessing to determine whether they represent valid multi-label cases or require correction. A distribution analysis of labels, where multi-label images are split into individual label counts, shows that "No Findings" is the most prevalent class, followed by Effusion, Infiltration, and Atelectasis. Hernia is the least common, highlighting class imbalance that will be addressed in model training to ensure robust performance across all categories.

Visualisations:

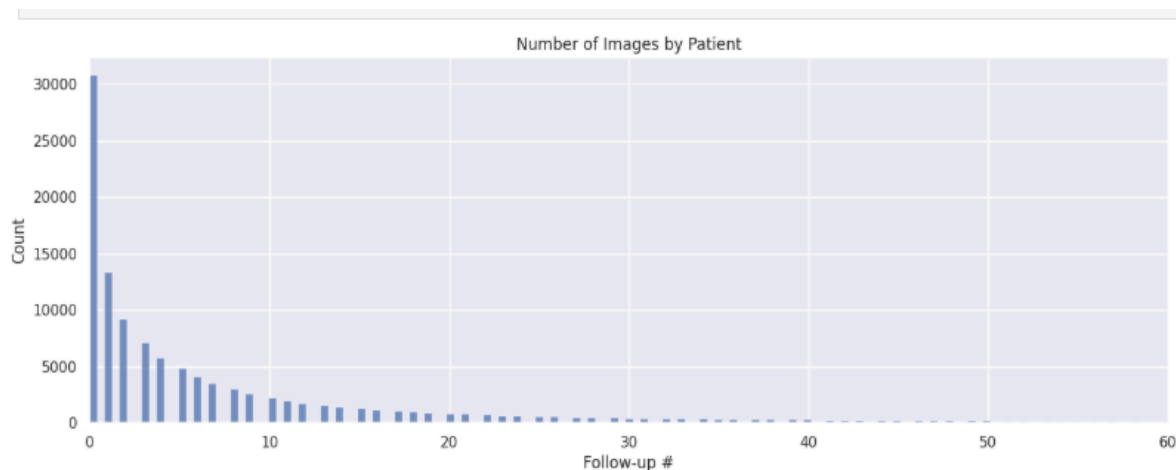


Figure 1 is an indicator of how many times a patient is seen. As expected, the plot is heavily right-skewed, and most patients have fewer than 10 images in the dataset. That being said, there are some extraneous outliers, such as the patient with 183 images in the dataset, as mentioned earlier.

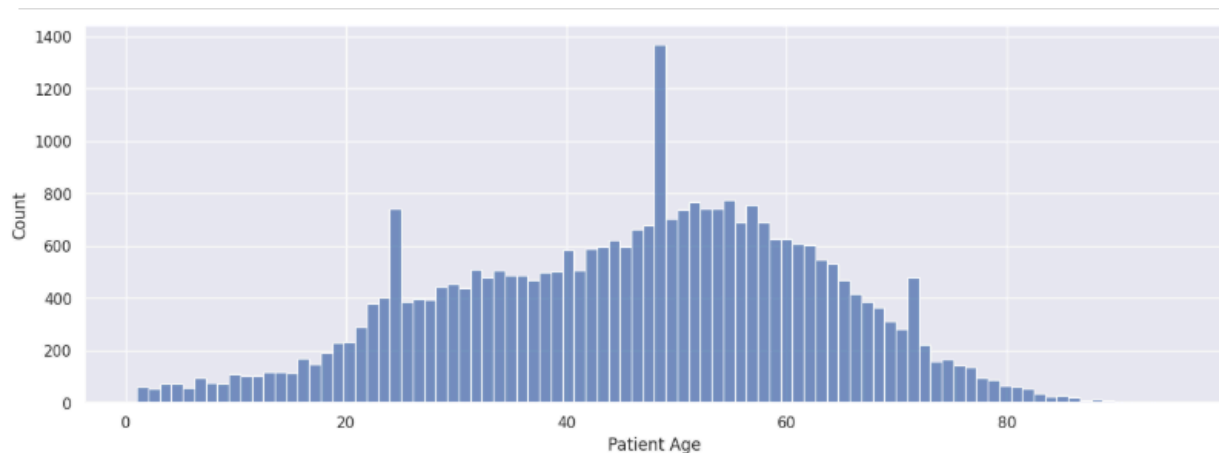


Figure 2. The patient age column contains several mislabeled entries, including one record indicating an age of 414 years, suggesting potential data entry errors or an unconventional scale. To address this, all records with a patient age exceeding 120 years were excluded from the dataset for analysis. The resulting distribution, visualised in the plot, shows that the majority of patients fall within the 30-70 age range, providing a robust representation of middle-aged adults. Additionally, the dataset exhibits a diverse range of patient ages, enhancing its applicability across different demographics.

Notably, three distinct groups display a significantly higher number of samples compared to their neighbouring age brackets, raising questions about the underlying data collection strategy. These peaks could indicate targeted data

gathering efforts or routine medical check-ups scheduled at specific ages, warranting further investigation into the dataset's acquisition process.

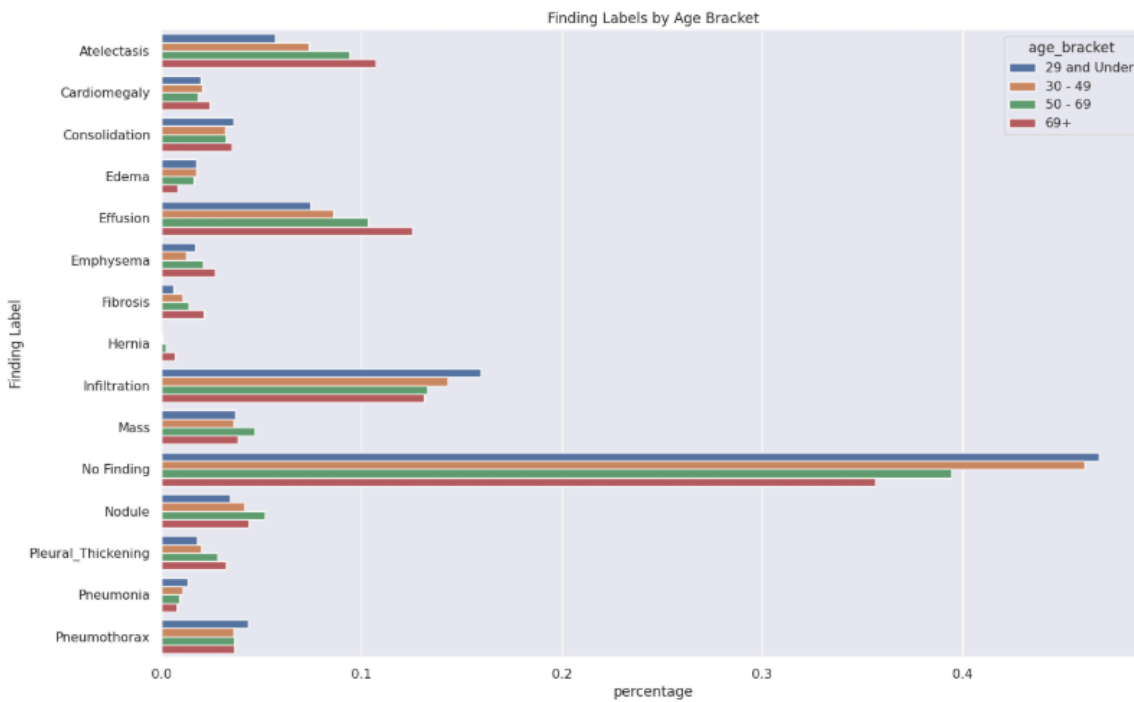


Figure 3. The distribution of finding labels across different age groups reveals notable variations, as illustrated in the accompanying plot. Patients under 29 years old exhibit a higher likelihood of being diagnosed with Pneumothorax, Infiltration, or "No Finding" compared to other age groups, suggesting a lower prevalence of severe conditions in younger individuals. In contrast, patients over 69 years old are more frequently diagnosed with Pleural Thickening, Fibrosis, Hernia, Emphysema, Effusion, Cardiomegaly, and Atelectasis. This increased prevalence may be attributed to older patients being less likely to have "No Findings" on their X-rays, reflecting a higher incidence of age-related conditions.

Patients in the 30-49 age group consistently show moderate rates across most labels, but they rank as a close second in the Edema and Cardiomegaly categories, indicating a notable presence of these conditions. Lastly, the 50-69 age group demonstrates a higher propensity for findings labelled as Nodule and Mass compared to other age groups, pointing to a potential age-related trend in these diagnoses.

Challenges

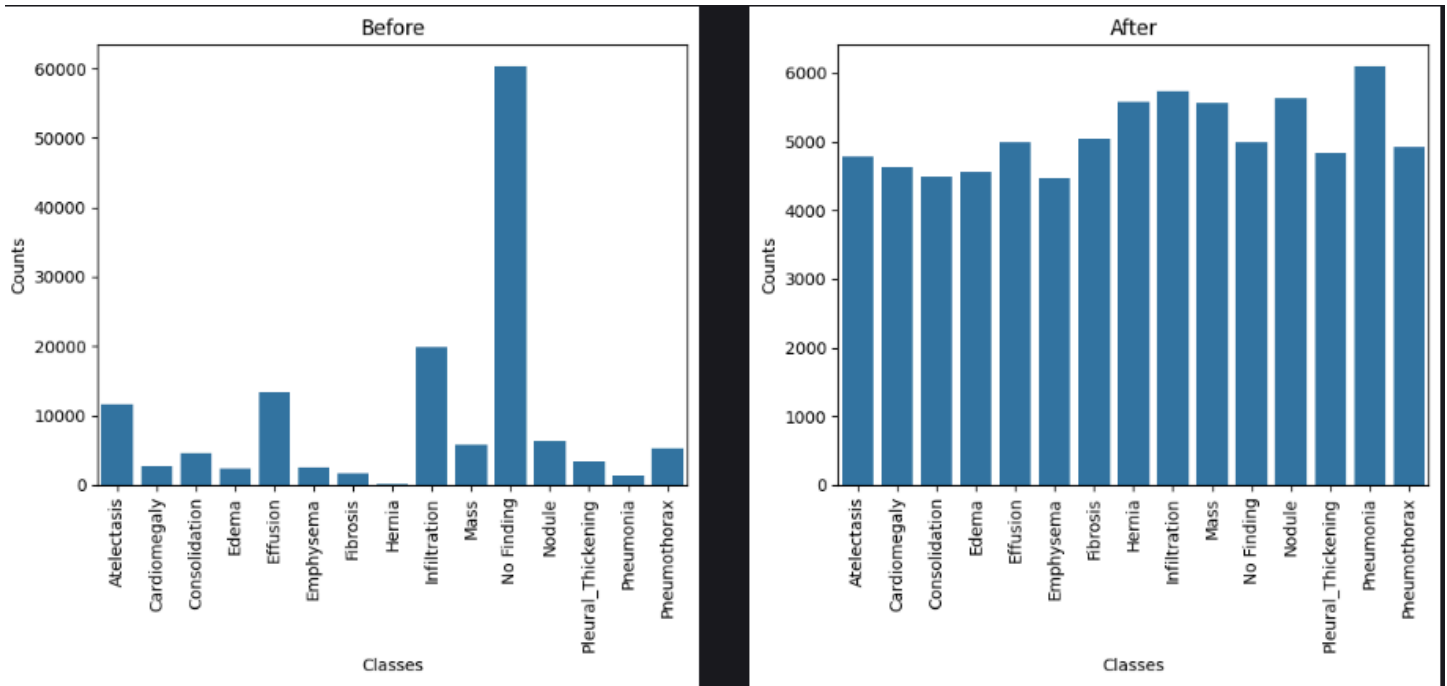


Figure 4. Despite the dataset's large size of 112,104 images, significant label imbalance was observed, as highlighted in prior visualisations. To address this, a revised version of the dataset was adopted, which offers a more balanced distribution of labels. However, this adjusted dataset contains only half the number of images (56,052), which may impact the model's training capacity but improves overall class representation, as seen in the updated distribution plot.

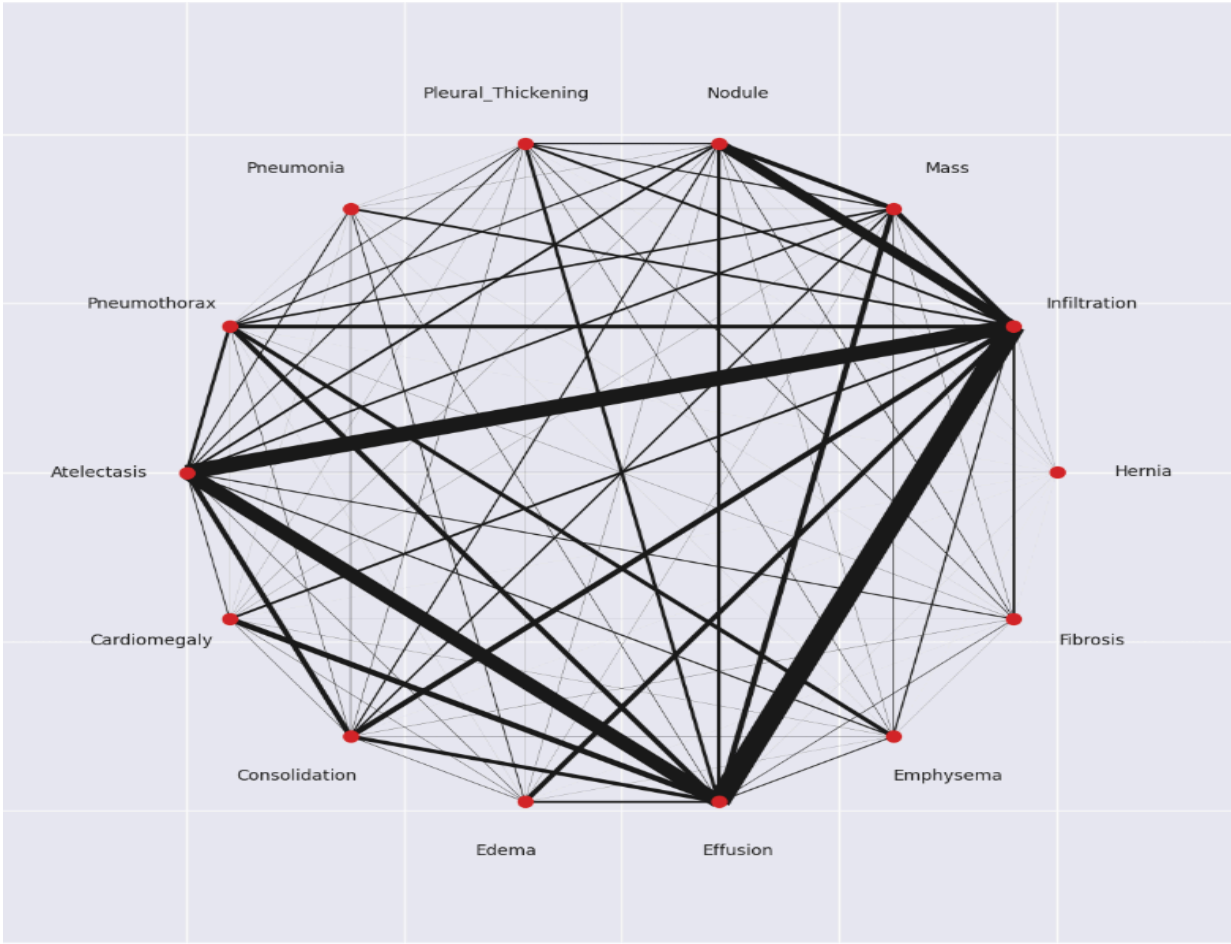


Figure 5. Label co-occurrence presents another challenge, as many images exhibit multiple diagnoses. This is visualized in two figures: a chord diagram where thicker lines indicate a higher likelihood of two labels occurring together (e.g., Infiltration and Effusion show strong co-occurrence), and a bar plot illustrating the distribution of multi-label occurrences within single images. These patterns necessitate careful handling in model development to accurately capture inter-label relationships.

Data Preparation

The NIH ChestX-ray14 dataset was used for this study. Images of patients over 120 years of age were excluded. The label field, which may contain multiple labels per image, was binarised using the MultiLabelBinarizer. The data was then divided into training (70%), validation (10%), and testing (20%) sets. Images were resized to either 224×224 or 256×256 pixels, depending on the requirements of the model, and subsequently normalised.

Models

After finishing our understanding and preparation of the data, this section focuses on the models’ architectures, the differences in AUC, and the interpretability of said models. Furthermore, this section will be split into 5 different categories depending on the architecture and methods used in each model.

Machine learning

Kernel SVM Approach

Following the approach outlined in Khan and Mehmood (2022) [5], a Kernel Support Vector Machine (SVM) framework was implemented to classify thoracic diseases using chest X-ray images from the ChestX-Ray14 dataset.

Data Preprocessing

A sampled subset (10%) of the ChestX-Ray14 dataset was utilized, comprising 51,000 frontal-view X-ray images from 836 patients, labeled with 14 thoracic diseases. Images were resized from their original resolution to 128x128 pixels to reduce computational demands while preserving key features. Pixel values were normalized to the range [0, 1] to ensure consistent input scaling for the model. To address high-dimensionality, Fast Principal Component Analysis (FPCA) was applied, reducing the 49,152 features (after flattening the images) to 100 principal components. FPCA enhances computational efficiency and mitigates outliers, aligning with the strategy for handling large-scale image data as described in [5].

SVM with Kernels

Four kernel types were implemented for the SVM model: Linear (L-SVM), Polynomial (P-SVM) with degree=3, Radial Basis Function (RBF-SVM), and Sigmoid (H-SVM). The dataset was split into 80% training and 20% testing sets. Each SVM model was trained with its respective kernel, setting the regularization parameter ($C = 1.0$) and enabling probability estimates to facilitate interpretability. Performance was evaluated using accuracy and classification reports, including precision, recall, and F1-score for each kernel, consistent with the evaluation metrics in [5].

Model Interpretability

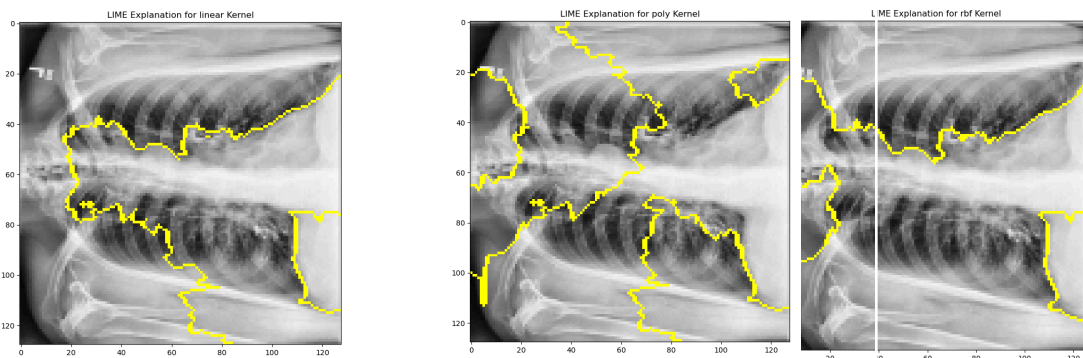


Fig. 6, Lime on three kernels

To enhance trust and validation in a medical context, Local Interpretable Model-agnostic Explanations (LIME) were employed. LIME visualizes regions of the X-ray images that most influence predictions, providing insights into the model's decision-making process.

Results and Challenges

The Linear, Polynomial, and RBF kernels achieved an accuracy of approximately 89.7%, while the Sigmoid kernel scored 87.3%. Classification reports revealed strong performance on the majority class (label "0"), but poor recall and precision for the minority class (label "1"), indicating potential data imbalance or a focus on binary classification of a single disease. LIME highlighted image regions contributing to predictions, enabling verification of clinical relevance, such as lung fields for pneumonia detection.

Challenges included high dimensionality, even after resizing, which necessitated FPC, though optimal component selection remained complex. Kernel parameter tuning was computationally intensive, and the sampled dataset underrepresented rare diseases, skewing performance. Training on large datasets demanded significant resources, mitigated by sampling but limiting scalability, echoing limitations noted in [5].

CNN and Transfer Learning

In this section, we will be focusing on 5 models, with the First being.

Fine-tuned DenseNet-121

Used DenseNet-121 as the backbone for the multi-label classification task due to its efficient feature reuse and strong performance on medical imaging.

Architecture

DenseNet-121, initialized with ImageNet weights, processes $256 \times 256 \times 3$ chest X-ray images through an initial convolution and pooling, four dense blocks (6, 12, 24, 16 bottlenecks), and finally a sigmoid classification layer [4].

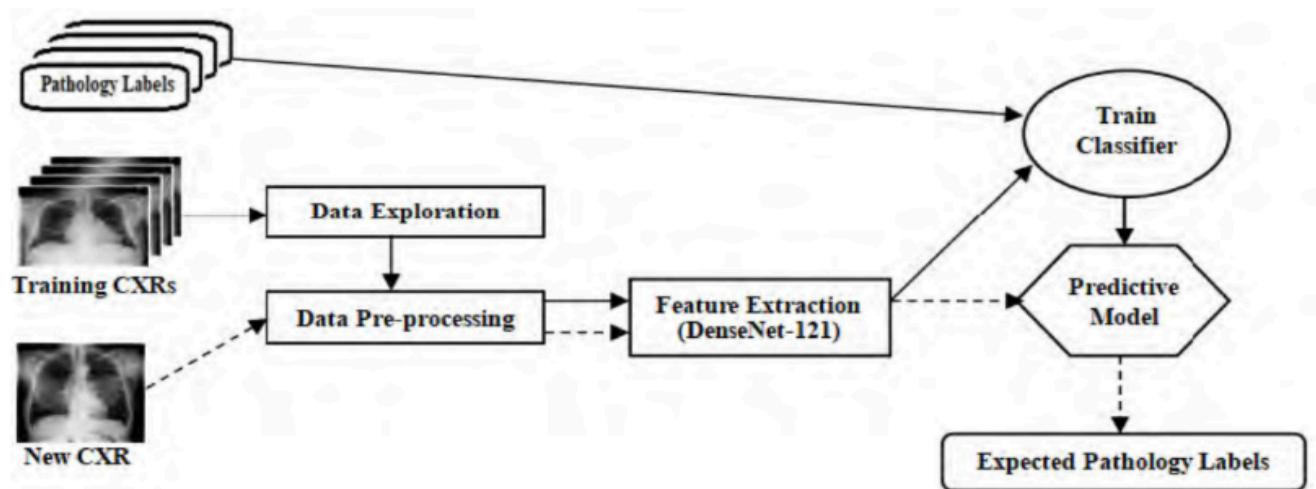


Fig 7, Model architecture

Fine-Tuning

We first fine-tuned all layers using Adam ($\text{lr} = 1 \times 10^{-3}$, $\text{batch} = 8$) and binary cross-entropy for up to 110 epochs with step-down lr scheduling on validation plateau. We then froze the backbone, extracted features, and trained a linear classifier for multi-label prediction [4].

Interpretability Techniques

We applied three interpretability methods in a single pipeline: Grad-CAM, class-specific heatmaps by back-propagating the predicted score into the final convolutional layer, guided backpropagation to derive saliency maps directly from input gradients, and LIME to fit simple local surrogate models on superpixel perturbations, highlighting the most influential regions for each label. Together, these techniques offer both global and local visual explanations of the model’s decisions on chest X-rays.

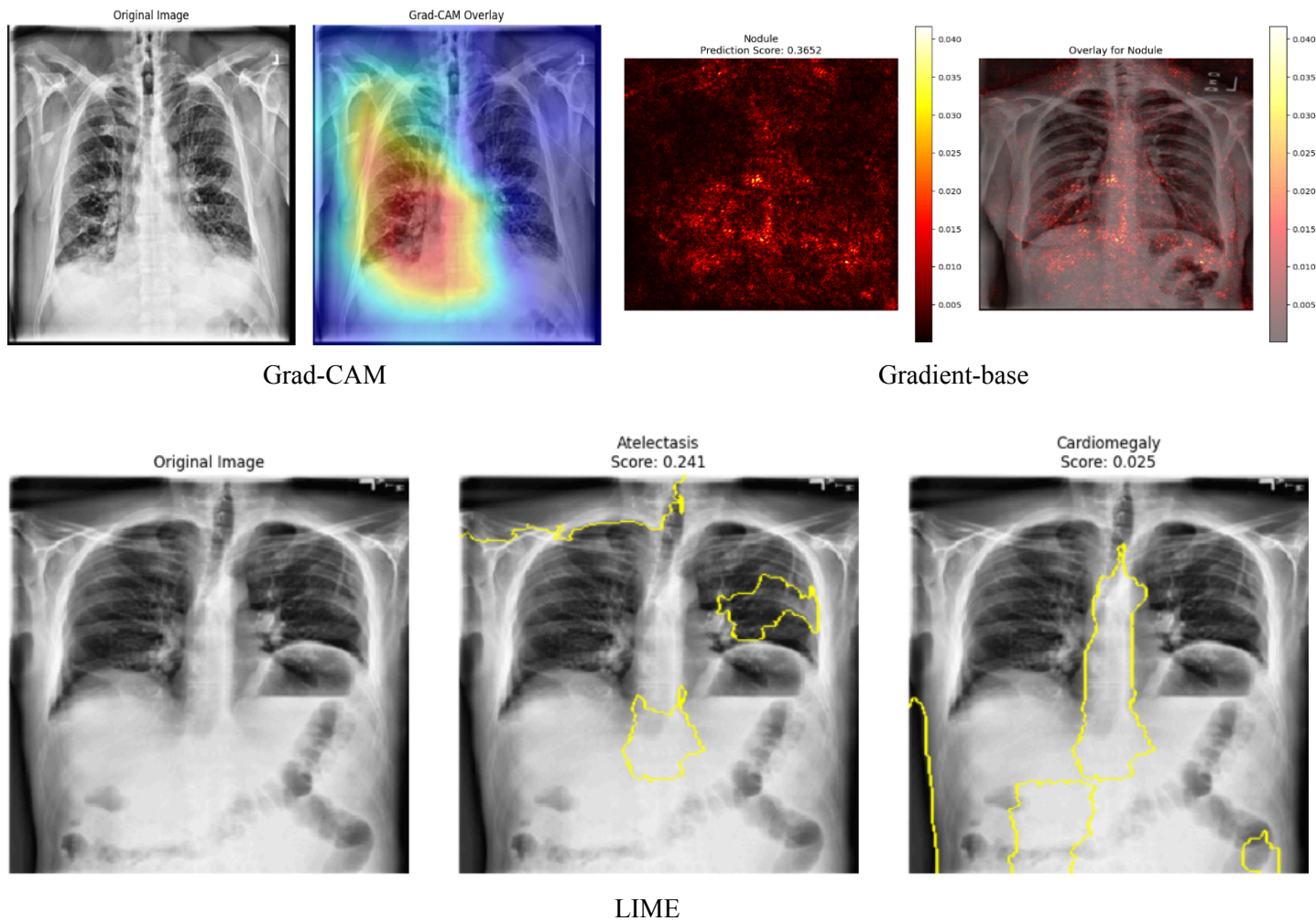


Fig 8, XAI techniques.

MobileNetV2(MobileLungNetV2)

Used MobileNetV2 as the backbone due to its lightweight and performance on medical images.

Architecture

MobileNetV2 as a backbone (ImageNet-pretrained, top layers removed) with a configurable input shape $224 \times 224 \times 3$ and append a lightweight head: a 3×3 Conv2D(32, ReLU) for feature refinement, 2×2 AveragePooling, Dropout(0.3), Flatten, and a final Dense layer with softmax over 15 classes. The resulting Model object cleanly separates the frozen feature extractor from our custom classifier layers [6].

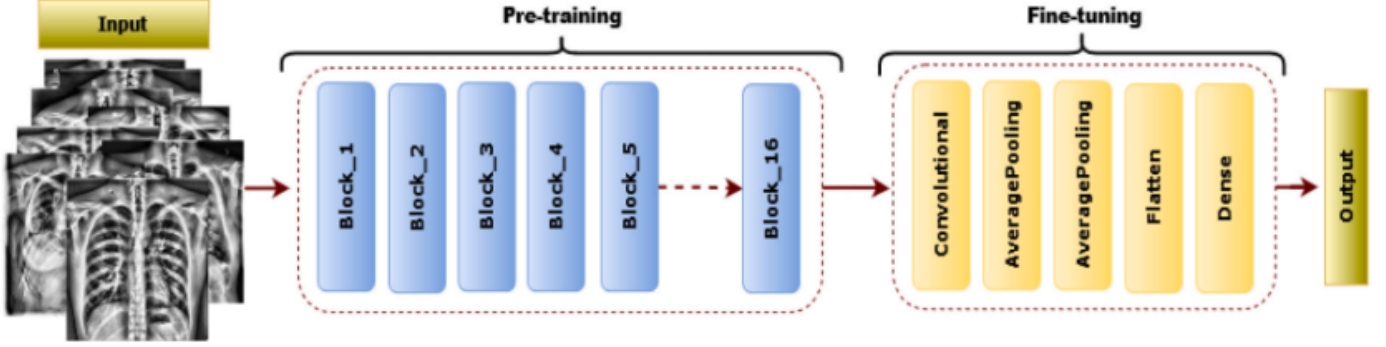


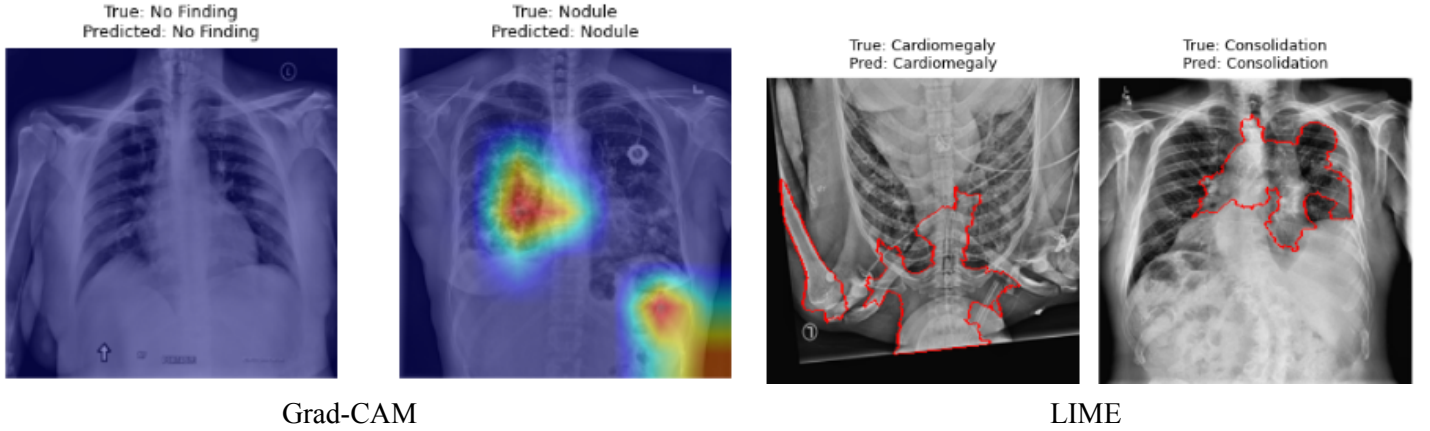
Fig 9, MobileLungNetV2 architecture

Fine-Tuning

We freeze a percentage of the backbone layers, 90%, and leave the last 10% trainable so that only the deepest MobileNetV2 blocks are fine-tuned alongside the new head. The compiled model uses RMSprop at $lr = 1e-3$, categorical cross-entropy loss, and tracks accuracy, precision, recall, and multi-label AUC. Training proceeds for up to 35 epochs with a batch size of 32 [6].

Interpretability Techniques

We applied four interpretability techniques to understand the model's decision and prediction. Grad-CAM makes a colour map over the X-ray by sending the predicted score back into the last feature layer, showing the rough areas the model cares about. Eigen-CAM finds the strongest pattern in those feature maps and highlights it without using gradients. LIME cuts the image into small patches, and fits a simple line model to see which patches change the prediction the most. SHAP uses ideas from game theory to give each pixel or patch a score for how much it helps or hurts the final decision. See Figure 10 for visualisations.



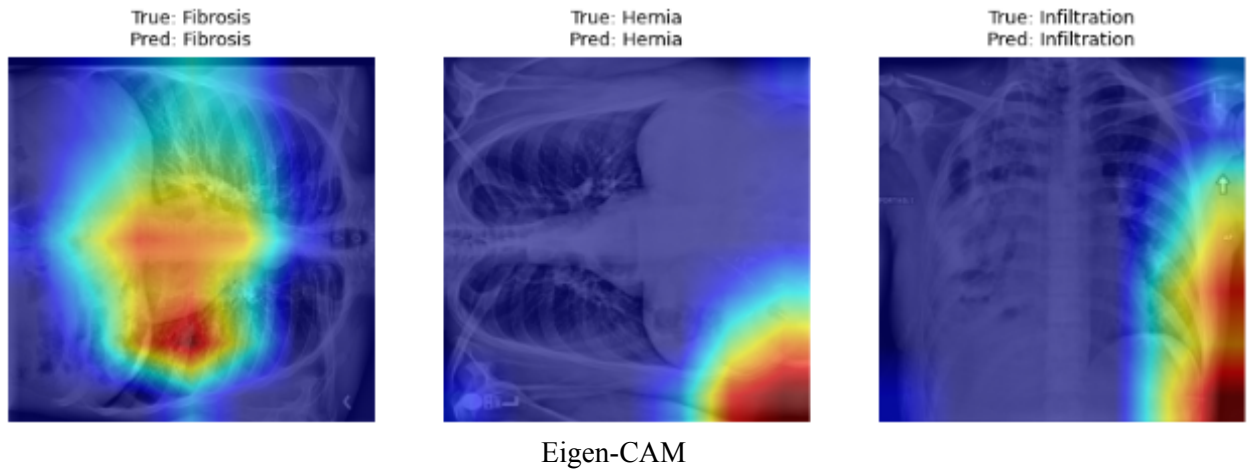


Figure 10, MobileLungNet: Explainable techniques.

Fine-tuned MobileNetV2

Used MobileNetV2 as the backbone due to its lightweight and performance on medical images.

Architecture

MobileNetV2 is composed of 17 lightweight blocks, each with a 1×1 expansion, a depthwise 3×3 convolution, and a 1×1 projection plus residual shortcuts, batch normalisation, and ReLU activations. The backbone is capped by a 1×1 convolution, global average pooling, and a final classification layer, delivering high accuracy with very low parameters and computational power [7]. The full architecture can be seen in Figure 11.

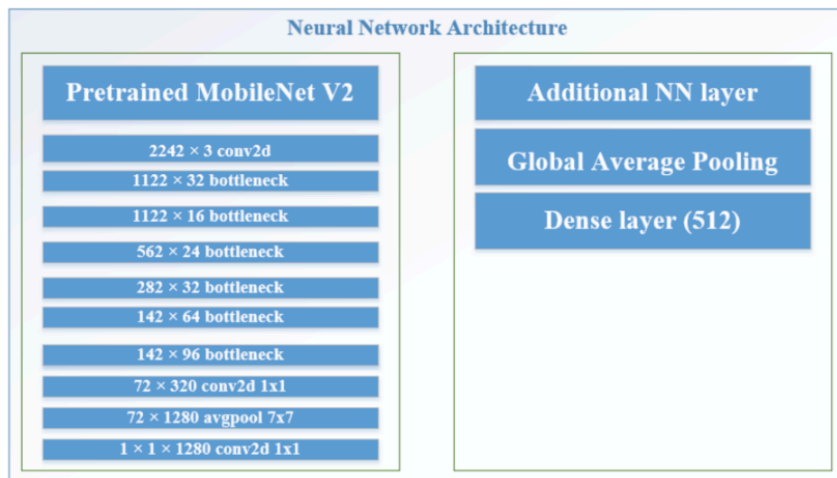


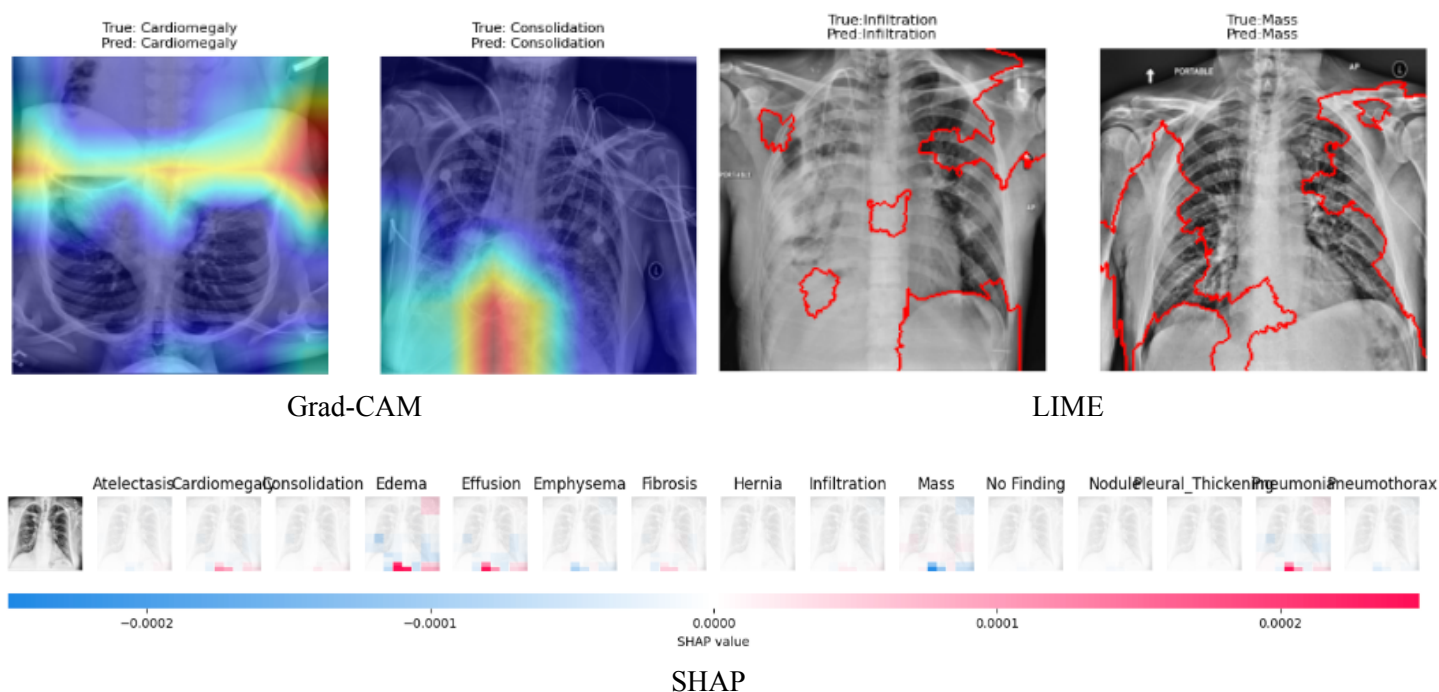
Fig 11. MobileNetV2 architecture.

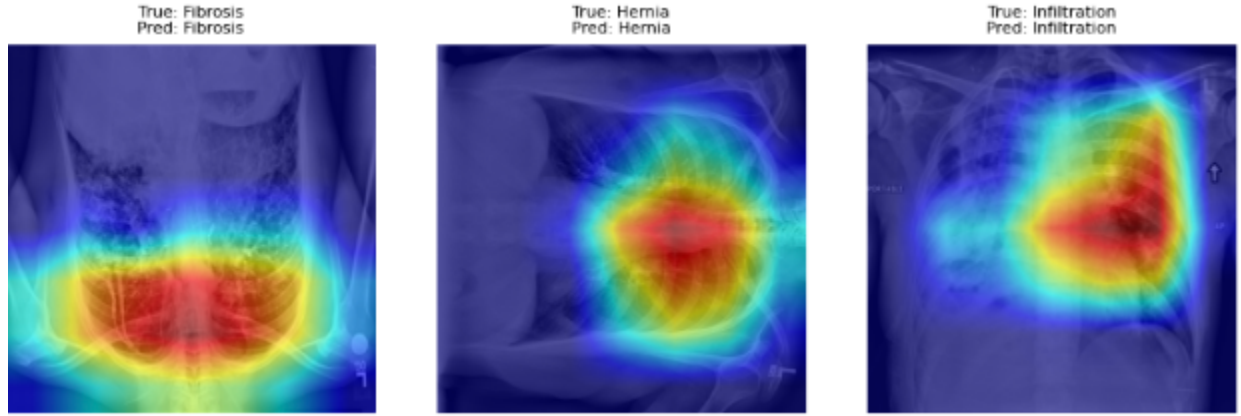
Fine-Tuning

We adapt MobileNetV2 to our multi-label chest X-ray task through a three-stage fine-tuning pipeline. In Phase I, we freeze the entire pre-trained backbone and train only the newly added global-pooling + sigmoid head for up to 10 epochs with a relatively high learning rate, monitoring validation loss and using early stopping (patience = 3) to prevent overfitting. Once the head has converged, Phase II unfetters the backbone weights and continues training all layers at a reduced learning rate for up to 15 additional epochs. In Phase III, we lower the learning rate once more and restore the original steps-per-epoch schedule, running up to 10 further epochs with a higher early-stopping patience (10) to finely adjust both backbone and head. Throughout, we use Adam optimization, binary cross-entropy loss, and track accuracy, AUC, precision, and recall on held-out validation data [7].

Interpretability Techniques

We applied four interpretability techniques to understand the model's decision and prediction. Grad-CAM makes a colour map over the X-ray by sending the predicted score back into the last feature layer, showing the rough areas the model cares about. Eigen-CAM finds the strongest pattern in those feature maps and highlights it without using gradients. LIME cuts the image into small patches, and fits a simple line model to see which patches change the prediction the most. SHAP uses ideas from game theory to give each pixel or patch a score for how much it helps or hurts the final decision. Check Figure 12 for visualisations.





Eigen-CAM

Figure 12, MobileNetV2 explainable techniques.

Dense Convolutional Network

Data Preprocessing

Data preparation involved merging Data_Entry_2017.csv and new_labels.csv using the "Image Index" column, extracted from image paths. The merged dataframe was filtered to include only the "Image Index," "Patient ID," and the 14 disease labels. Images were normalised using ImageDataGenerator with sample-wise centring and standard normalisation. For training, augmentation included a rotation range of 10 degrees and a width shift range of 0.1. Images were resized to 320x320 pixels to maintain feature integrity while accommodating the DenseNet121 architecture. To prevent data leakage, patient-level splitting was performed, dividing unique patient IDs into 70% training (21,628 images), 15% validation (4,689 images), and 15% test (4,509 images) sets using a 42 random seed.

Model Selection and Architecture

The DenseNet121 model was selected due to its proven efficacy in medical imaging tasks, as demonstrated by CheXNet [3], which outperformed radiologists in pneumonia detection (AUC = 0.896). The model leveraged transfer learning from ImageNet weights, with the first 150 layers frozen to retain pre-trained features. A GlobalAveragePooling2D layer followed by a Dense layer with a sigmoid activation was added to output probabilities for the 14 labels. This architecture aligns with the multi-label classification approach outlined in [1], addressing the limitations of single-label frameworks noted in earlier studies [1][2].

Hyperparameter Tuning and Optimization

The model was compiled with the Adam optimizer, setting the learning rate to $1e-4$ to balance convergence and stability. A custom weighted cross-entropy loss function was implemented to address class imbalance, calculated as $(L_{cross-entropy}^w = -(w_p y \log(f(x)) + w_n (1 - y) \log(1 - f(x))))$, where (w_p) and (w_n) , which are weights derived from positive and negative frequencies in the training set. Class weights were computed by dividing the sum of each label by the total number of samples, ensuring underrepresented classes contributed more to the loss. Training spanned 8 epochs with a batch size of 16, monitored by an EarlyStopping callback on validation loss with a patience of 3, optimizing for computational efficiency and model generalization.

Explainability Techniques

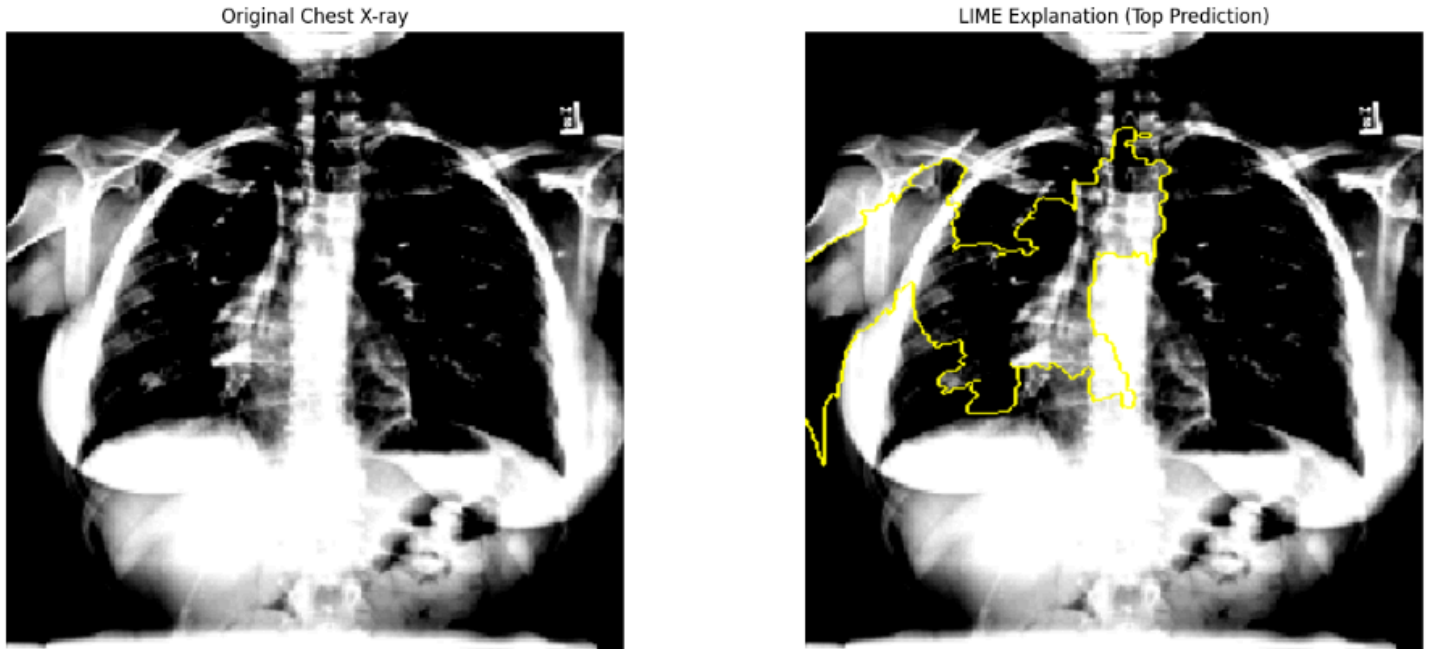


Fig 13. Lime explanation on the top predicted class.

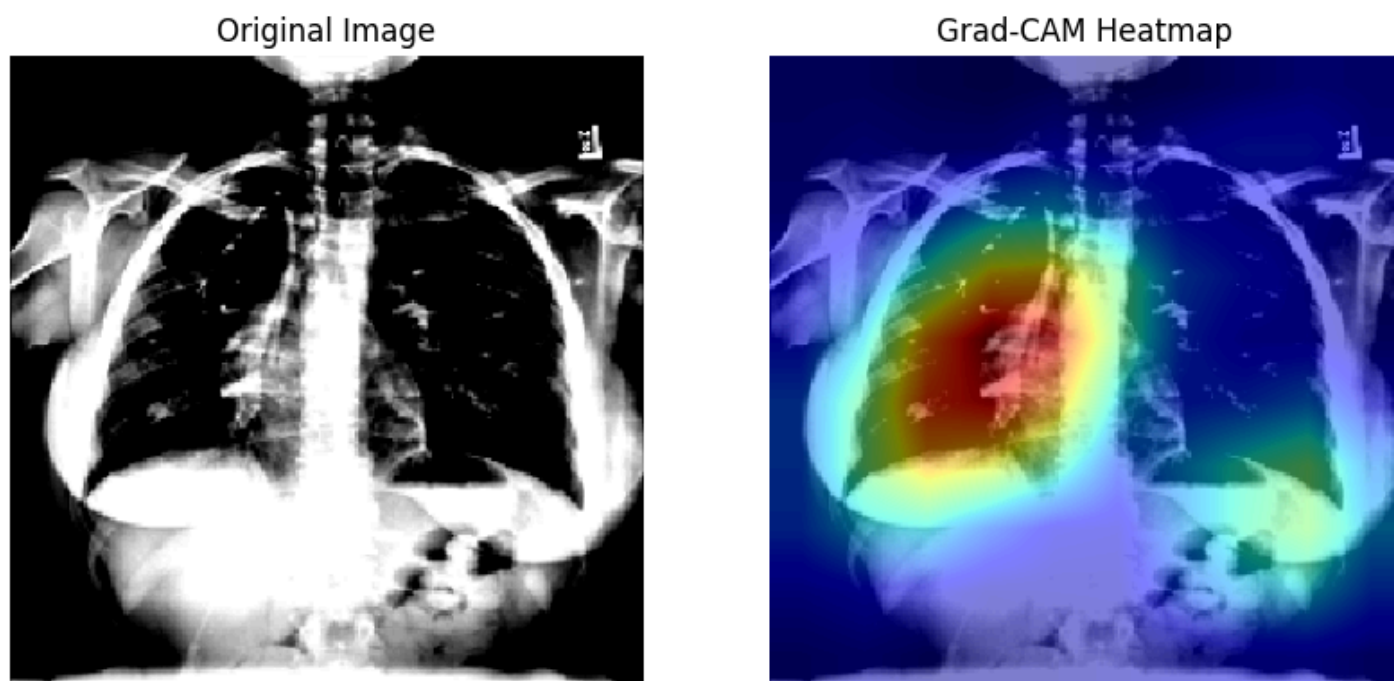


Fig 14. Grad-CAM on the same image as in **Fig. 13**.

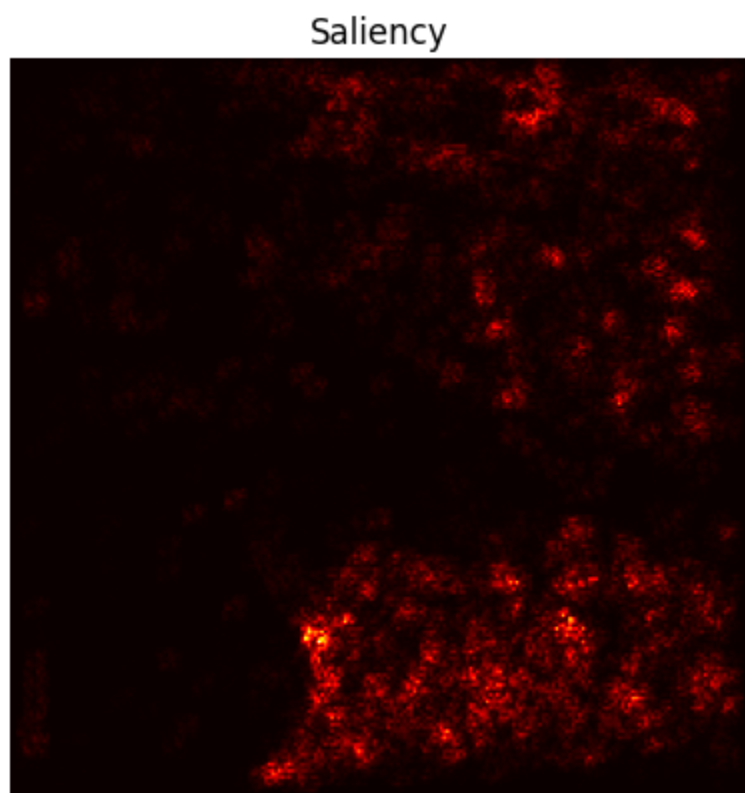


Fig 15. Saliency map on **Fig. 13**.

Model interpretability was enhanced using Grad-CAM, LIME, and saliency maps. Grad-CAM localized decision-making by computing gradients of the target output for feature maps, visualized as heatmaps. LIME provided local explanations by perturbing input images and observing prediction changes. Saliency maps, generated using `tf.GradientTape` highlights

pixel-level influences on predictions, offering insights into disease-specific regions. These methods, consistent with [1], improved trust and clinical validation.

Evaluation Methods

Model performance was assessed using the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) for each of the 14 pathologies on the test set. Predictions were generated with the model. Predictions and AUC scores were computed using `roc_auc_score` from `scikit-learn`, reflecting the multi-label nature of the task. This metric, widely adopted in [3][6][7], provided a robust measure of classification performance across imbalanced classes.

Comparison with Alternatives

While single-label models like GoogleNet [2] and ensemble CNNs [6][7] achieved high AUCs (0.964 and 0.99, respectively) on smaller datasets, DenseNet121's multi-label capability addressed co-occurring disease detection, a limitation in ChestX-ray8 [4][5]. Comparative studies [1] confirm DenseNet-based models' superiority, with the proposed method achieving competitive AUCs (e.g., 0.90 for Cardiomegaly).

CustomCNN for Multi-Label Thoracic Disease Classification

To explore baseline performance using a simple yet effective approach, we implemented a Custom Convolutional Neural Network (CustomCNN) trained from scratch for multi-label classification. This model was designed to classify the 14 thoracic diseases in the ChestX-ray14 dataset using resized grayscale images.

Architecture

The architecture consists of three convolutional layers with ReLU activation, each followed by a max pooling layer to reduce spatial dimensions. After feature extraction, the output is flattened and passed through two fully connected layers, ending with a sigmoid activation to support multi-label output. Dropout layers were added between the dense layers to prevent overfitting. Refer to Figure 16 for the model architecture.

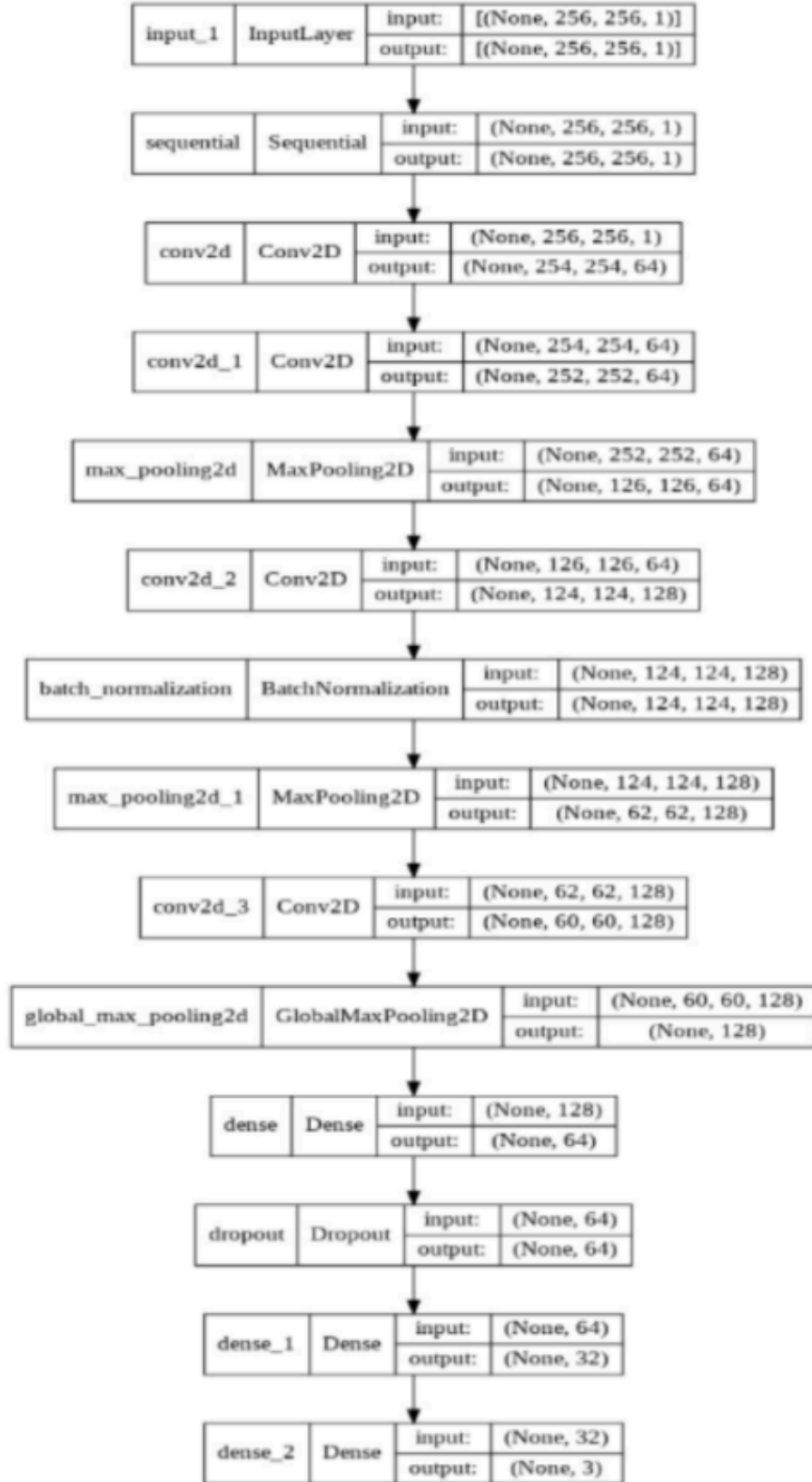


Fig. 16. Model architecture

Training Setup

The model was trained on images resized to 224×224 pixels. Binary cross-entropy was used as the loss function with the Adam optimizer. The training was conducted over 30 epochs with early stopping based on validation loss, and a learning rate scheduler was applied to reduce the learning rate on the plateau.

Interpretability Techniques

To interpret the predictions of the model, four techniques were applied. **Grad-CAM**, as seen in Figure 17, highlights the most influential areas of the X-ray image by visualising gradients flowing into the final convolutional layer. **LIME**, as seen in Figure 18, provides superpixel-based visualisations indicating the contribution of each region to the model's output. **SHAP**, as seen in Figure 19, uses cooperative game theory to assign importance scores to image regions, explains model confidence for each class. Finally, **DeepDream**, as seen in Figure 20, visualises patterns that activate different neurons by enhancing the input image in the direction of increasing neuron activation.

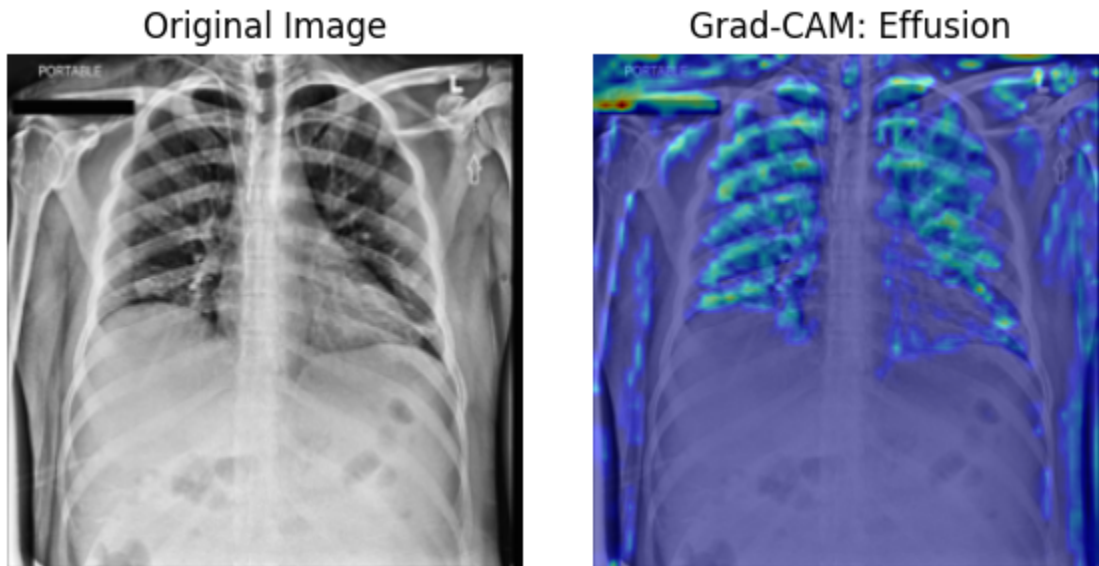


Fig 17. Grad-CAM

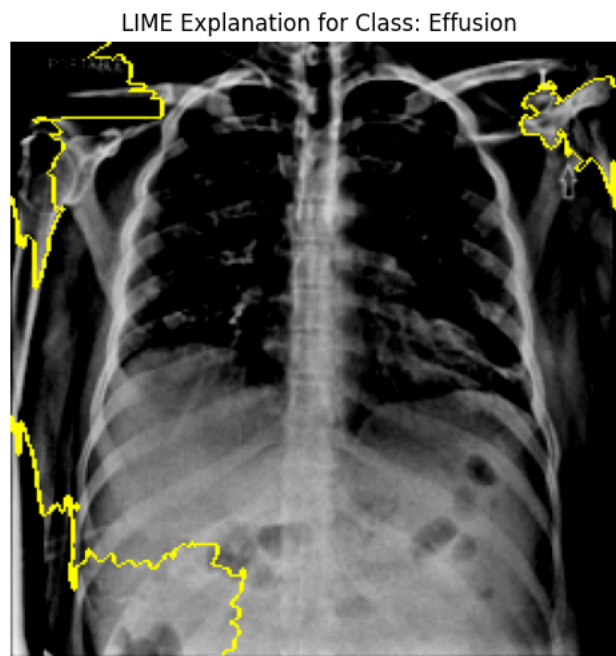


Fig 18. LIME

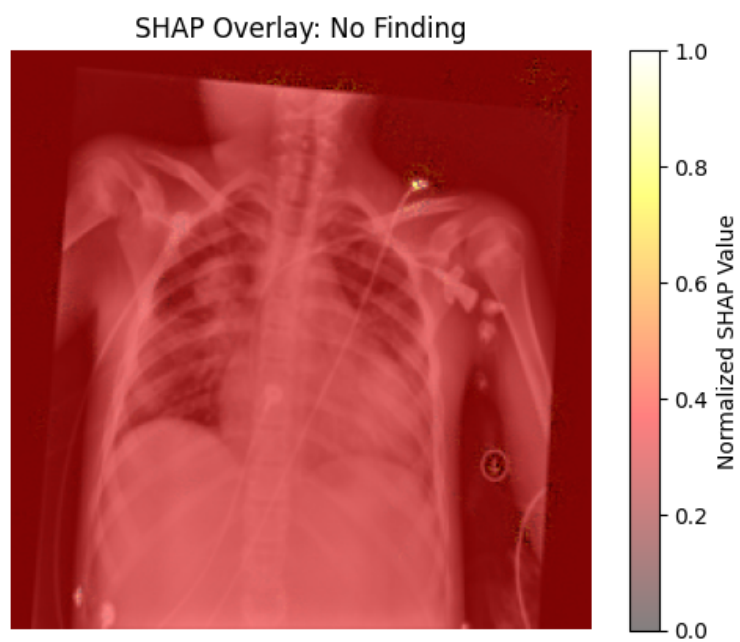


Fig 19. SHAP

DeepDream Visualization - CustomCNN (Grayscale)

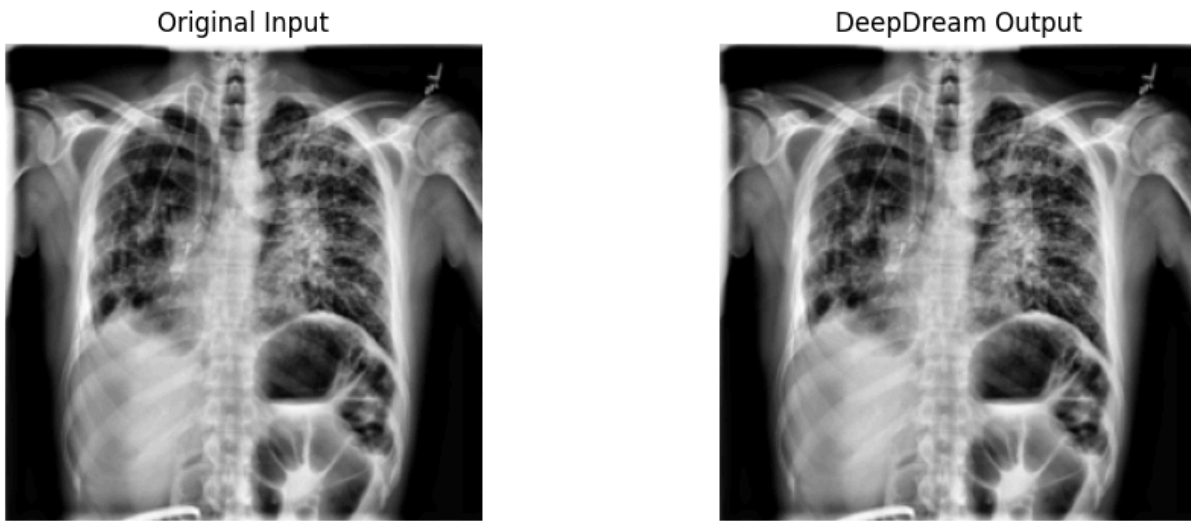


Fig 20. DeepDream

Transformers

SwinChex

In this section, we will be focusing on the SwinChex Transformer[18], which uses a Swin Transformer [19] for multi-label chest X-ray classification. Vision Transformers (ViTs), though originally designed for natural language processing, have shown strong performance in computer vision tasks. However, they face several challenges when applied directly to visual data.

In contrast to textual tokens, visual elements in images exhibit significant scale variation, which presents a challenge for fixed-size patch embeddings commonly used in standard Vision Transformers (ViTs). This scale discrepancy is especially critical in vision tasks such as object detection and semantic segmentation, where precise spatial details are essential. Additionally, the high resolution of images leads to computational limitations, as traditional self-attention mechanisms in ViTs demonstrate quadratic complexity to image size. This inefficiency renders them unsuitable for processing large-scale medical images effectively. Swin Transformers, and by extension SwinCheX, were proposed to deal with the mentioned challenges and operate as a general-purpose backbone for various vision tasks.

Architecture

The model Architecture can be seen in Figure 21.

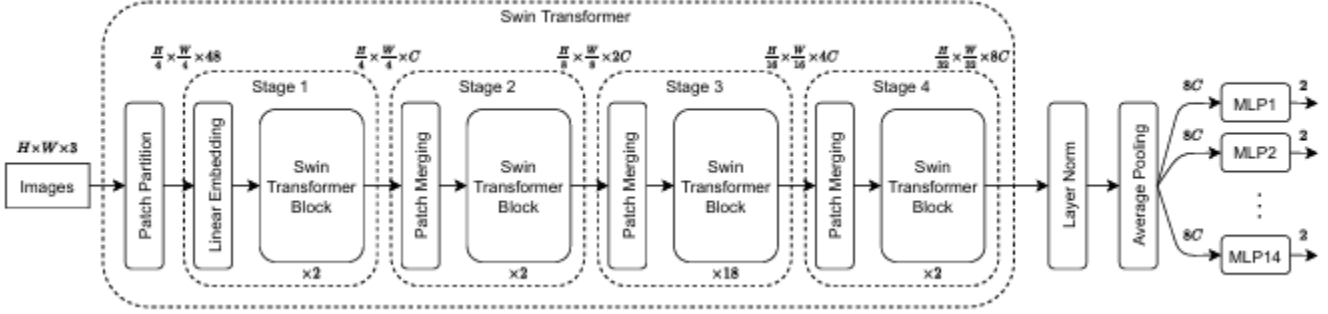


Fig 21. Chest X-ray images, after being converted to RGB and resized to 224, are passed through a Swin-L transformer. Then that is followed by a Layer Norm and an average pooling of 7×7 . Finally, it is passed through 14 Multi-Layer Perceptrons(MLP), referring to the 14 different diseases.

Fine-Tuning

To train the model for this task, binary cross-entropy with logits (BCEWithLogitsLoss) was used as the loss function. This formulation is particularly suited for multi-label problems, as it treats each class prediction as an independent binary classification task and ensures numerical stability by internally combining the sigmoid activation with the cross-entropy loss.

Model fine-tuning was conducted using the Hugging Face Trainer API, which facilitates integrated training, evaluation, and checkpoints. The training consisted of three epochs, with a batch size of 32 for both training and evaluation phases. An initial learning rate warm-up was implemented over 500 steps, and weight decay was set to 0.01 to prevent overfitting. The evaluation and checkpointing strategies were epoch-based, allowing performance metrics to be assessed consistently at the end of each training cycle. The best-performing model was selected automatically using validation performance as the criterion. All training logs were recorded at intervals of ten steps, and model output and logs were stored in designated directories.

Comparison

The created model produced an AUC score of 0.795 compared to the original paper's 0.81 due to reduced epochs for time constraints. Furthermore, the original paper used Grad-CAM for visualisation, as can be seen in Figure 22. However, our paper used Lime for visualising, which can be seen in Figure 23.

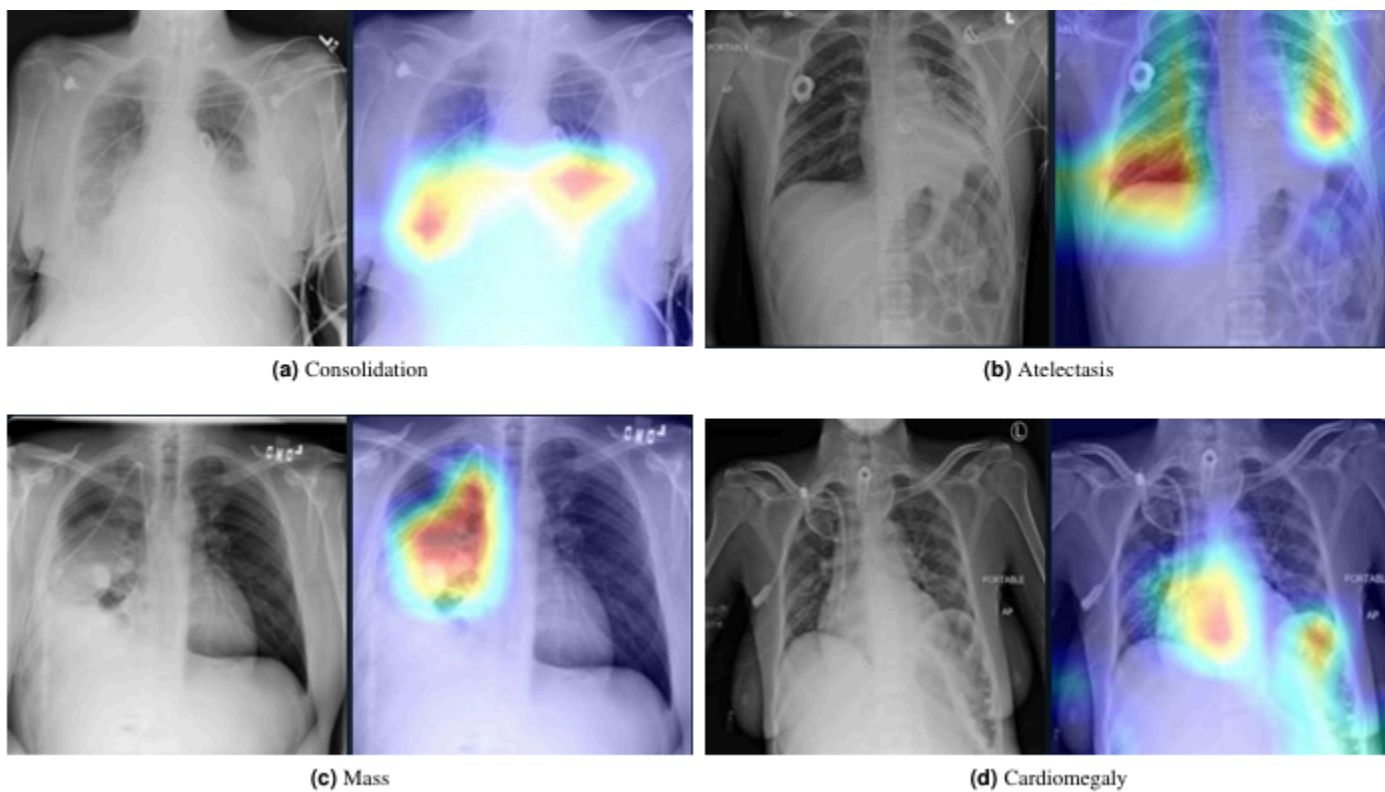


Figure 22. Grad-Cam of the Paper for samples containing Consolidation, Atelectasis, Mass, or Cardiomegaly.

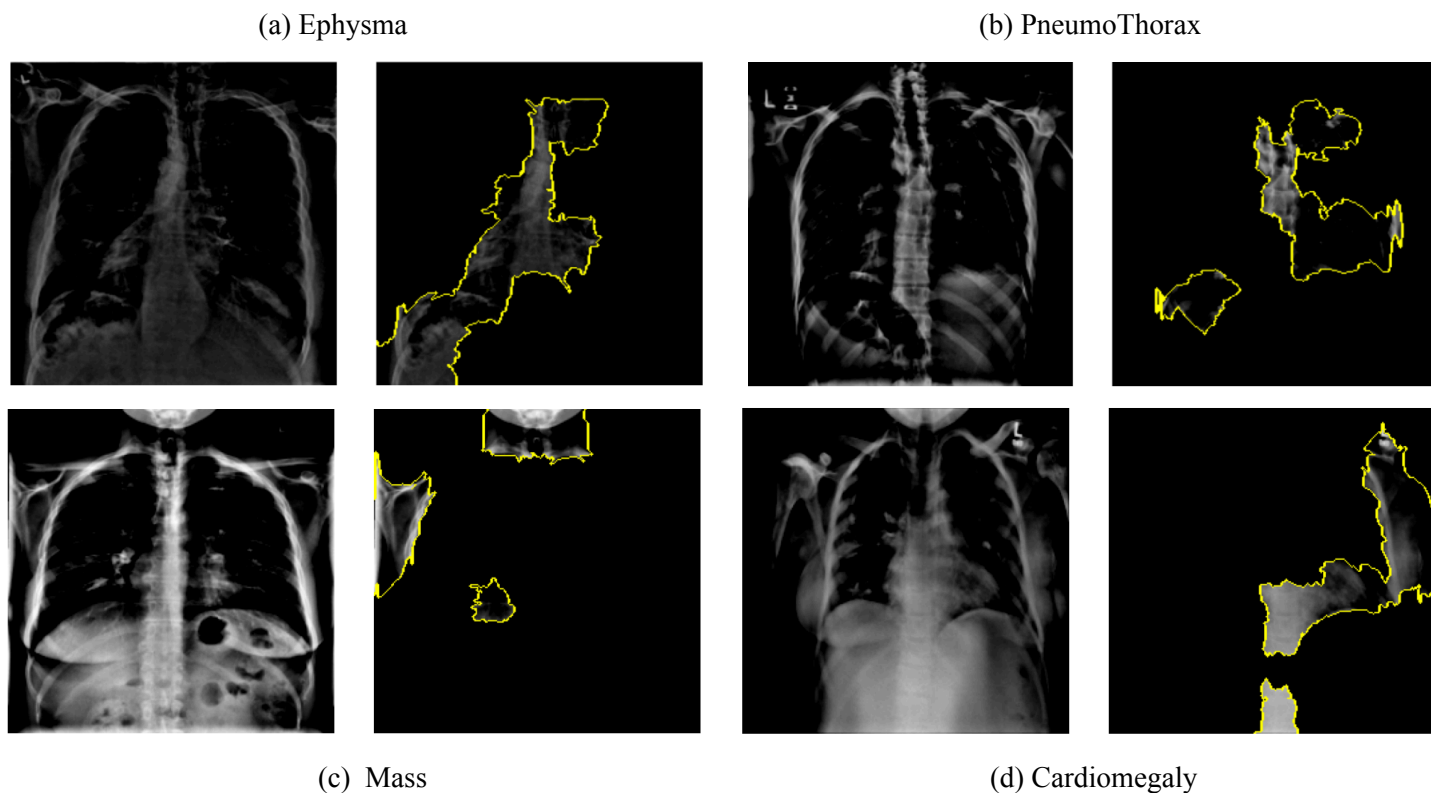


Figure 23. Lime of our Implementation for samples containing Ephysma, PneumoThorax, Mass, or Cardiomegaly

Auto Encoders Framework

In this section, we will be focusing on 2 AutoEncoder Inspired models

AE-CNN (Autoencoder + DenseNet121)

The first model is a custom AutoEncoder-based CNN designed to learn compressed feature representations while simultaneously performing disease classification. Unlike previous approaches that use pretrained encoders, this model jointly trains both the encoder-decoder architecture and the classification head from scratch.

Architecture

The encoder consists of several convolutional and pooling layers that reduce the image dimensions while capturing key features. The resulting latent feature representation is passed to two branches: a decoder network that reconstructs the input image for denoising, and a classifier composed of dense layers with sigmoid activation to predict disease labels.

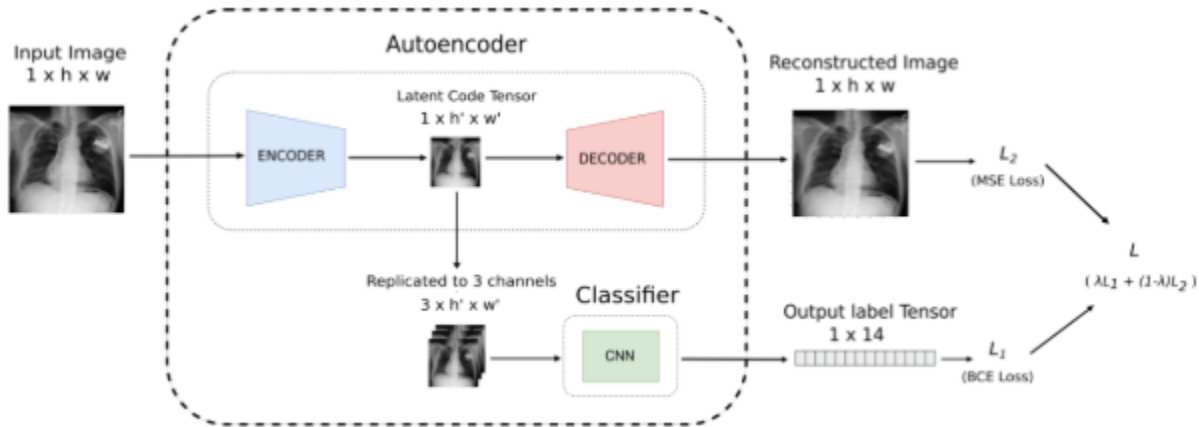


Fig 24. AE-CNN architecture

Training Setup

The model was trained on a subset of 1000 images from the ChestX-ray14 dataset due to runtime limitations. Images were resized to 224×224 pixels and normalized. The training was performed over 10 epochs using binary cross-entropy for the classification head and mean squared error (MSE) for the reconstruction loss.

Interpretability Techniques

Grad-CAM was applied to visualize the most influential regions of the chest X-ray that contributed to the model’s prediction by computing gradients of the predicted class concerning the final convolutional layer. The resulting heatmaps were overlaid on the original images, and both the predicted and true labels were displayed to facilitate direct visual comparison and assessment of model accuracy.

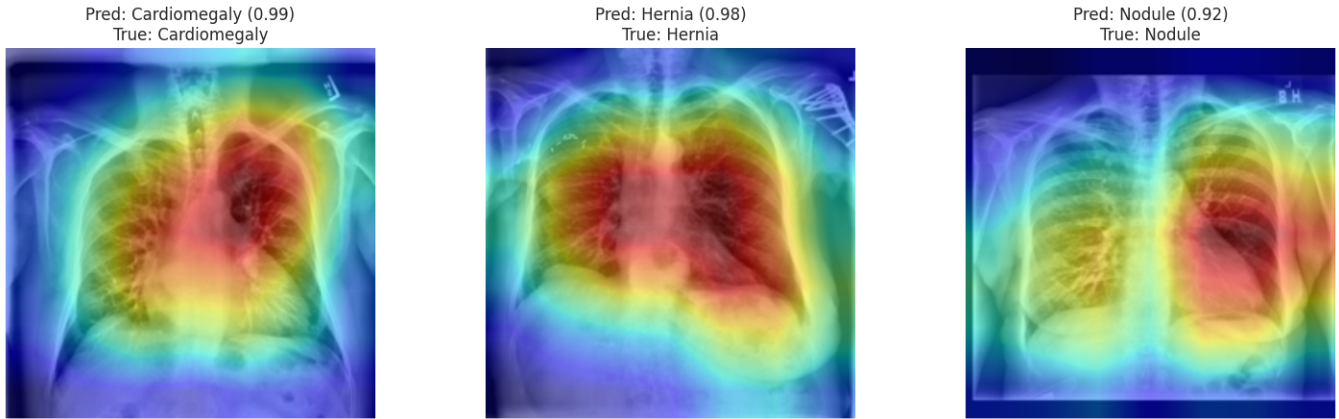


Fig 25. Grad-CAM

Multi-Level Residual Feature Fusion Network

The Second auto-encoder-based model is the **Multi-Level Residual Feature Fusion Network (MLRFNet)** [Citation]. Although previous methods use attention mechanisms to guide the model’s focus on key features, they rely solely on the output of the final feature map from a CNN during classification and lack the utilisation of semantic information at different levels. This limits the improvement of classification accuracy. The MLRFNet proposed the ECA attention mechanism and the MRFC module to enhance the model’s attention to critical features in channels and space, which is conducive to improving classification accuracy.

Architecture

The model Architecture can be seen in Figure X.

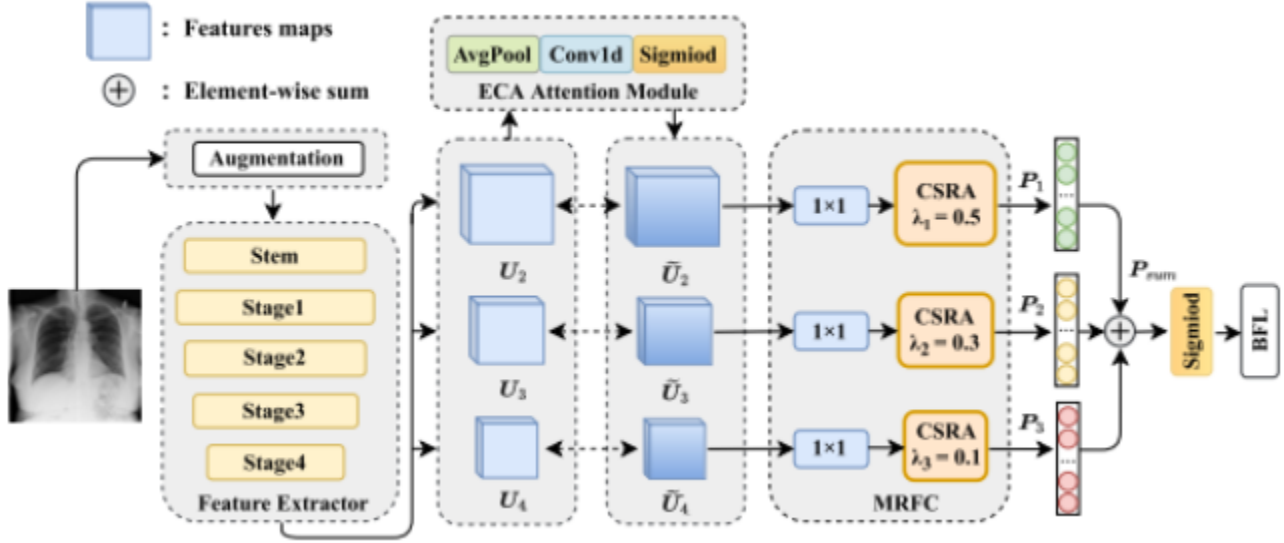


Fig 26. Reviews the entire MLRFNet architecture, which consists of feature extraction, ECA attention Modules, and a multi-level residual feature classifier(MRFC).

First, the feature extractor, which is based on the Res2Net50 backbone, is designed to enhance multi-scale feature representation through hierarchical residual connections. The input chest X-ray image is first processed by a 7×7 convolution followed by a 3×3 max pooling layer, producing the initial feature map U_0 . The network then progresses through four sequential stages, each comprising bottleneck blocks that combine standard convolutions with Res2Net modules. These stages reduce the spatial resolution while increasing the channel depth, yielding feature maps U_1 through U_4 . Specifically, Stage 1 applies three bottleneck units and outputs a $56 \times 56 \times 256$ tensor; Stage 2 applies four units to produce a $28 \times 28 \times 512$ map; Stage 3 uses six units for a $14 \times 14 \times 1024$ map; and Stage 4 includes three units producing a $7 \times 7 \times 2048$ tensor. These feature maps capture hierarchical semantic information at multiple scales, enabling the model to localise lesions of varying size and texture. The overall structure of the feature extractor is illustrated in **Figure Y**.

Layer	Output size	Operation	Feature map
Input	$224 \times 224 \times 3$		
Stem	$112 \times 112 \times 64$	Conv 7×7	U_0
Stage1	$56 \times 56 \times 64$	Max pool 3×3	U_1
	$56 \times 56 \times 256$	Conv 1×1 R2conv 3×3 Conv 1×1 $\times 3$	
Stage2	$28 \times 28 \times 512$	Conv 1×1 R2conv 3×3 Conv 1×1 $\times 4$	U_2
Stage3	$14 \times 14 \times 1024$	Conv 1×1 R2conv 3×3 Conv 1×1 $\times 6$	U_3
Stage4	$7 \times 7 \times 2048$	Conv 1×1 R2conv 3×3 Conv 1×1 $\times 3$	U_4

Fig 27. The Res2Net50 Architecture

Then we use the ECA attention module on the output feature maps, helping the network pay more attention to the relevant channel feature of the lesion. This will be done using the following equation.

$$\tilde{U} = U \odot \text{sigmoid}[C1D_k(\text{GAP}(U))], \tilde{U} \in \mathbb{R}^{H \times W \times C},$$

Finally, the model should consider both the image's global information and the spatial position of lesions on the feature map. In various stages of CNN's feature extraction, MRFC prioritises essential places of diseases and reduces irrelevant information. It can be calculated using Figure Z.

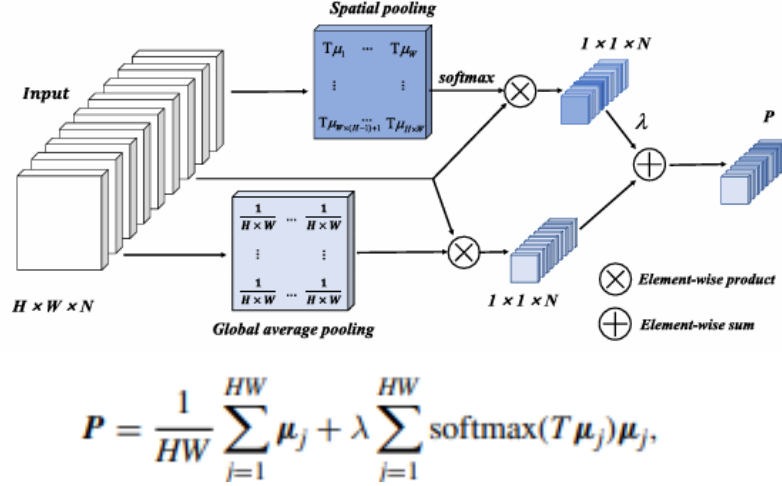


Fig 28. The MRFC calculation

Fine-Tuning

The MLRFNet model was fine-tuned using the Adam optimiser with a binary cross-entropy loss function. The model was trained over 20 epochs with early stopping to prevent overfitting. The training process used three key callbacks. First, a ModelCheckpoint was set to save the model weights corresponding to the highest validation AUC score, ensuring retention of the best-performing model. Second, a ReduceLROnPlateau scheduler monitored the validation loss and reduced the learning rate by 0.1 if no improvement occurred within three epochs. Third, early stopping was used with a patience of ten epochs, based on the validation AUC, and was set to restore the best model weights. We used AUC as the main metric during model evaluation.

Comparison

After the evaluation was done, our model got an AUC score of 0.519 compared to the original paper's 0.853, which was due to the model's complexity on our end, where there were issues during implementation, also our implementation is in TensorFlow compared to the original's PyTorch implementation. Furthermore, the original paper used Grad-CAM for visualisation, as can be seen in Figure A. However, our paper used Lime and Grad-Cam for visualising, which can be seen in Figure B.

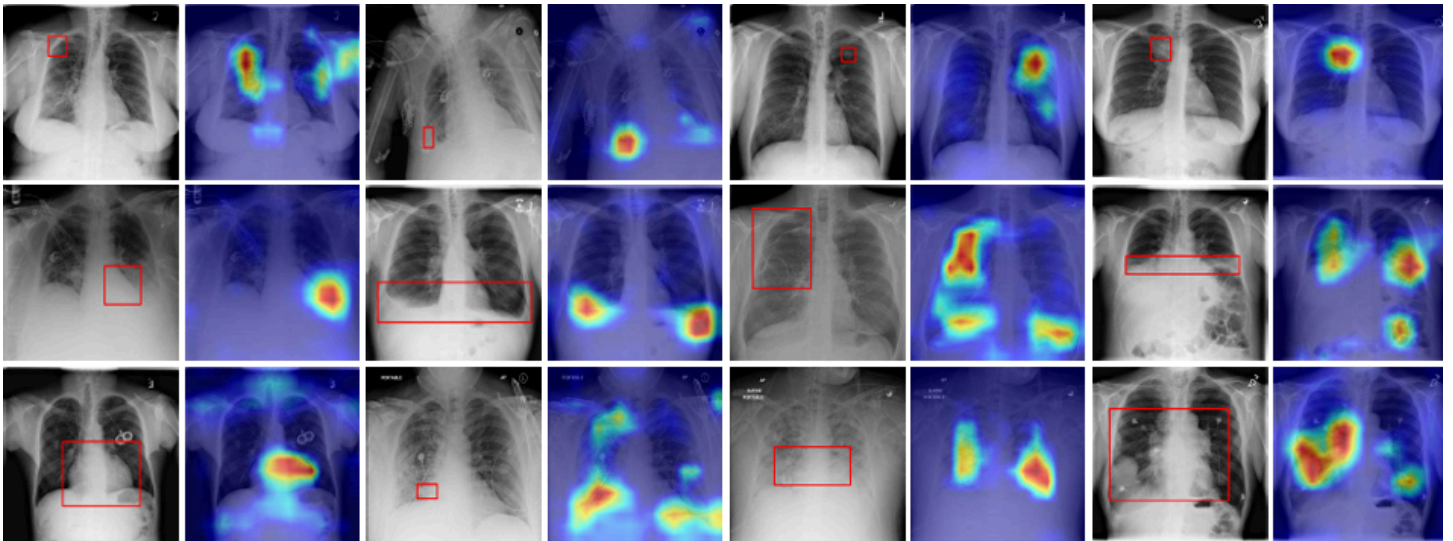


Fig 29. Original Paper's Visualization

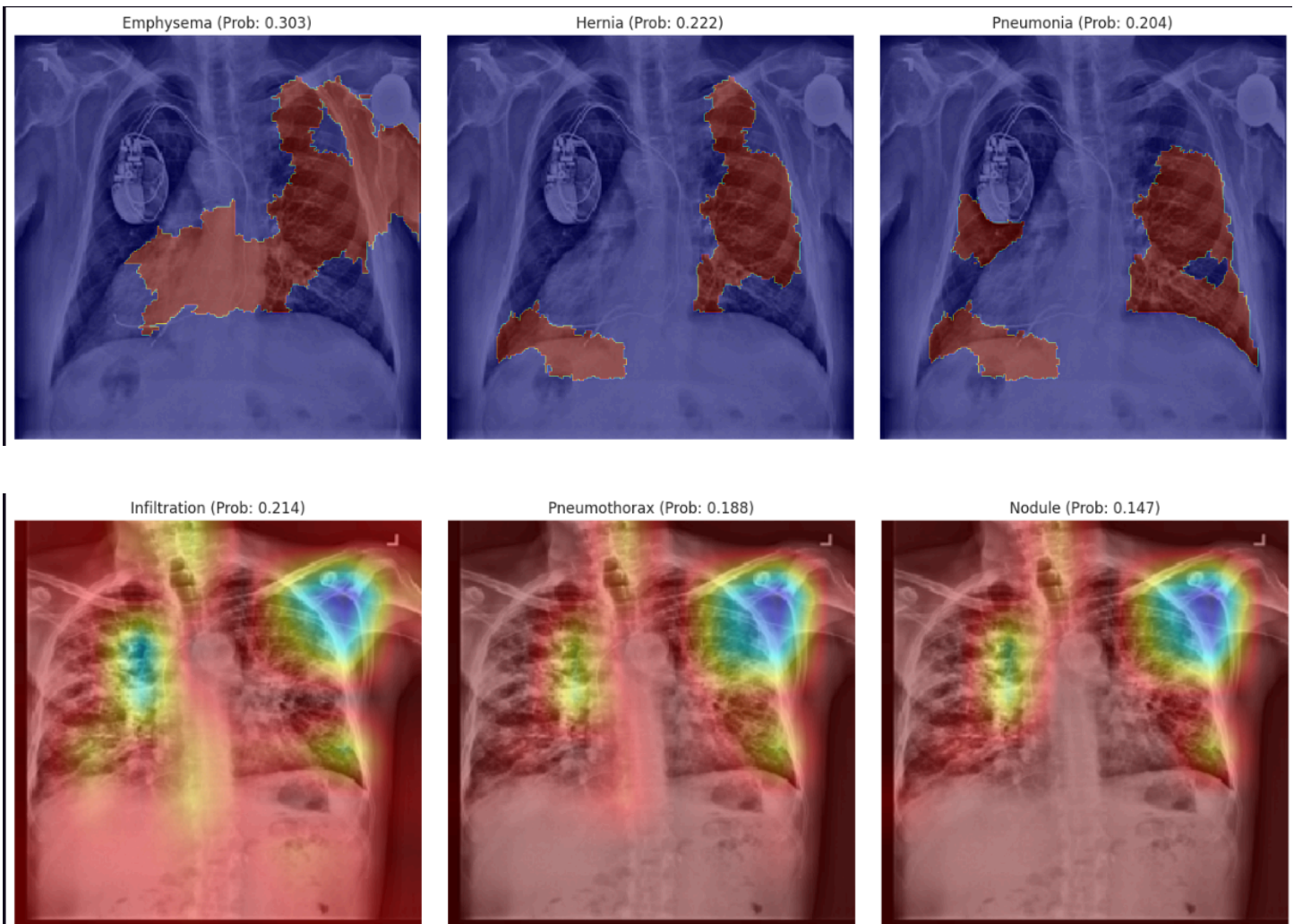


Fig 30. Our Paper

Attention Mechanism

This section includes models that leverage architectural designs to enhance feature focus during classification. The following Residual CNN model incorporates skip connections that support gradient flow and improve the network's ability to capture relevant features.

CRAL Variants (ResNet-50 / DenseNet-121 + Attention Blocks)

Architecture

The Residual CNN architecture is composed of five sequential residual blocks, each containing two convolutional layers with batch normalization and ReLU activation functions. To preserve gradient flow and mitigate vanishing gradients during training, skip connections are applied by adding the input of each block to its output. Following the residual blocks, a global average pooling layer is used to reduce the feature maps, and a final dense layer with sigmoid activation generates multi-label predictions across the 14 thoracic disease classes.

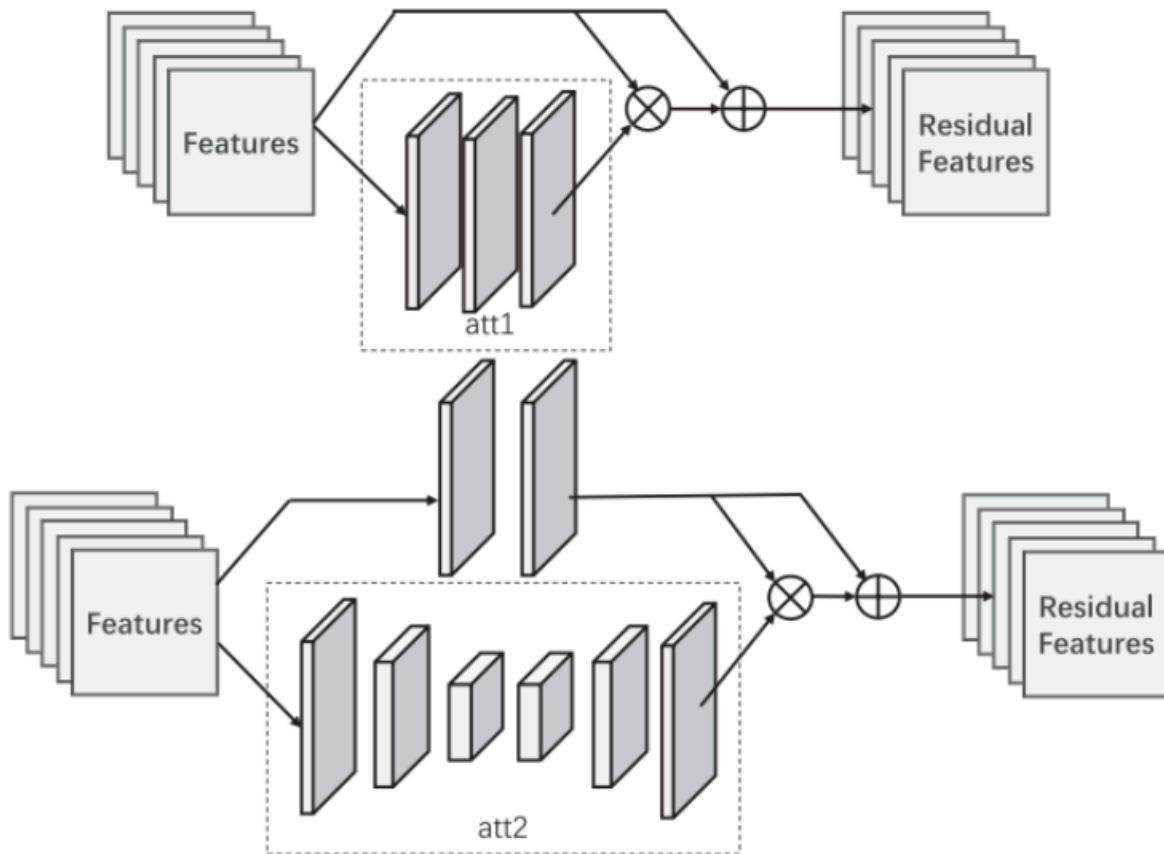


Fig 31. CRAL architecture

Training Setup

All images were resized to 224×224 pixels before training. The model was trained for a single epoch using the Adam optimizer and binary cross-entropy loss. Due to runtime limitations, the training duration was restricted, which impacted performance but allowed for evaluating the model's structure and interpretability techniques.

Interpretability Techniques

Four interpretability techniques were applied to analyze the model's predictions. Grad-CAM was used to highlight discriminative regions in the chest X-rays by computing gradients concerning the final convolutional layer, offering insight into how the residual connections influenced spatial focus. LIME provided local explanations by perturbing image regions and fitting interpretable models, while SHAP assigned feature importance scores based on cooperative game theory to assess global impact. Additionally, DeepDream was employed to visualize activation patterns by enhancing input features that maximally stimulated specific neurons, revealing abstract feature sensitivities within the network.

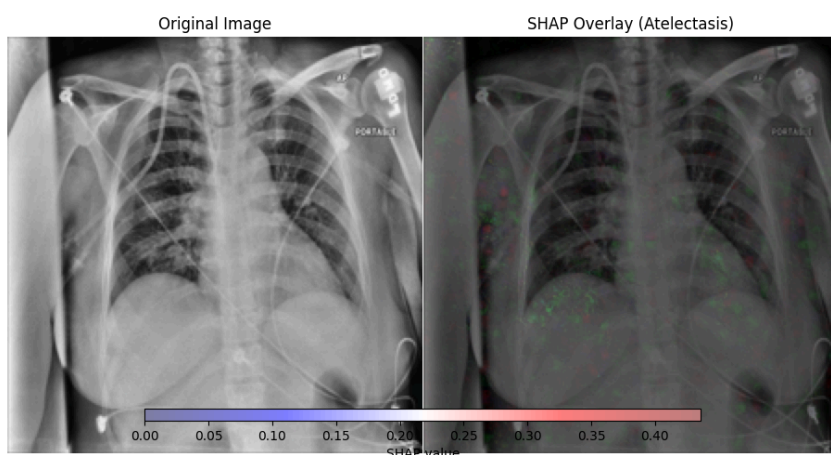


Fig 32. SHAP

Grad-CAM (Class 1)

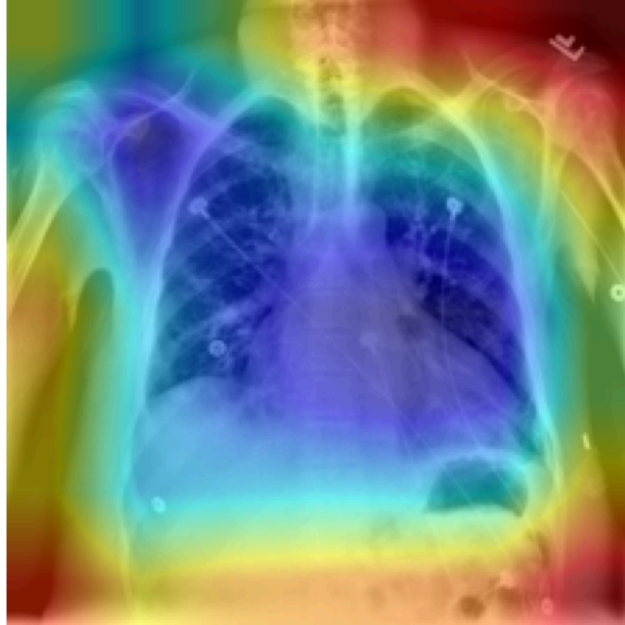


Fig 33. Grad-CAM

LIME Explanation (Class 10)

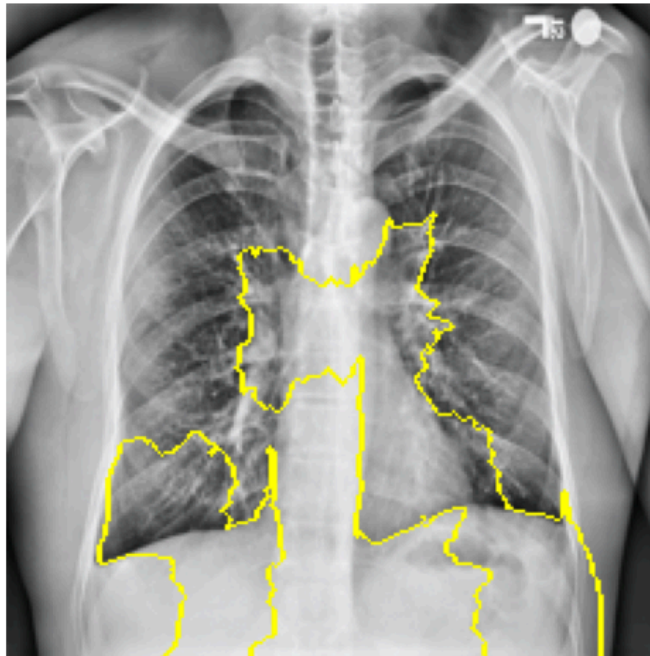


Fig 34. LIME

DeepDream: Original vs. Enhanced

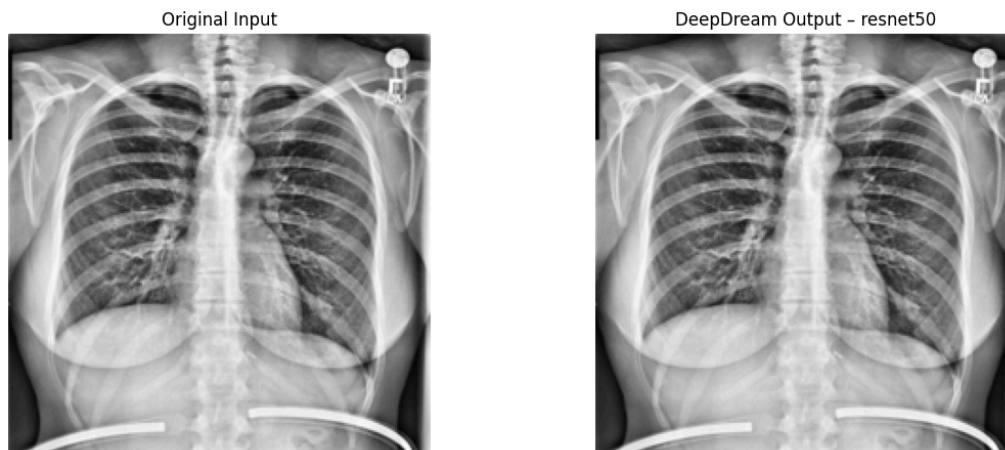


Fig 35. DeepDream

XAI

Explainable artificial intelligence (XAI) in deep learning-based medical image analysis

In this section, we will be focusing on 2 papers incorporating Explainability and Interpretability techniques for the Chestx-Ray14 Dataset. With the first paper being **Explainable artificial intelligence (XAI) in deep learning-based medical image analysis**

Data Preprocessing

Data preprocessing involved merging the label and metadata files, retaining "Image Index," "Patient ID," and the 14 disease labels. Images were loaded using TensorFlow, decoded as JPEG with three channels, and resized to 224x224 pixels. Normalisation was applied by dividing pixel values by 255.0. A patient-level split was performed with a 42 random seed, allocating 70% (approximately 35,760 images) for training, 15% (approximately 7,650 images) for validation, and 15% (approximately 7,590 images) for testing. The train_test_split function ensured no data leakage, with data batched into sets of 32 using tf.data.Dataset [1, Section 2.3].

Model Development

A DenseNet121 model was adapted for multi-label classification, utilising transfer learning with ImageNet weights. The architecture included an input tensor of shape (224, 224, 3), with the base model's top layer excluded. A GlobalAveragePooling2D layer, followed by a Dense layer with 512 units and ReLU activation, a Dropout layer (0.5 rate), and a final Dense layer with 14 units and sigmoid activation, was added. The model was compiled with the Adam optimiser and binary cross-entropy loss, trained for 8 epochs with early stopping on validation loss, monitored by an EarlyStopping callback [1, Section 2.4].

Explainability Techniques

Explainability was achieved using Grad-CAM, which localised model decisions by computing gradients of the target output relative to the conv5_block16_concat layer’s feature maps, producing heatmaps overlaid on original images. The method involved a gradient tape to capture activations and gradients, followed by global average pooling and normalisation. This approach, detailed in [1, Section 3.1.1], provided visual insights into disease-specific regions, enhancing clinical interpretability [1, Section 3.1.1.3].

Evaluation Methods

Model performance and explanation quality were evaluated using Dice and Intersection over Union (IoU) scores. Predictions from Grad-CAM heatmaps were thresholded at 0.3 and dilated with a 5x5 kernel to generate binary masks. Dice score measured overlap as $((2 \cdot \text{intersection}) / (\text{sum of predictions} + \text{sum of ground truth} + 1e - 8))$, while IoU was calculated as $(\text{intersection} / (\text{union} + 1e - 8))$. Results were logged to /kaggle/working/evaluation_metrics.txt, aligning with the computational assessment recommended in [1, Section 4.2].

Mask-free Medical Model Interpretation

The second paper is **Mask-free Medical Model Interpretation (MAMMI)**, which is created to address key challenges in applying neural concept annotation to the medical domain, including the need for a domain-specific concept set, the impact of class imbalance, and the limitations of general vision-language models. MAMMI tackles these issues by utilising a curated medical concept set and a vision-language model trained specifically on medical data, enabling effective and interpretable neuron annotation without relying on pixel-wise segmentation masks.

Architecture

The model architecture can be seen in Figure X.

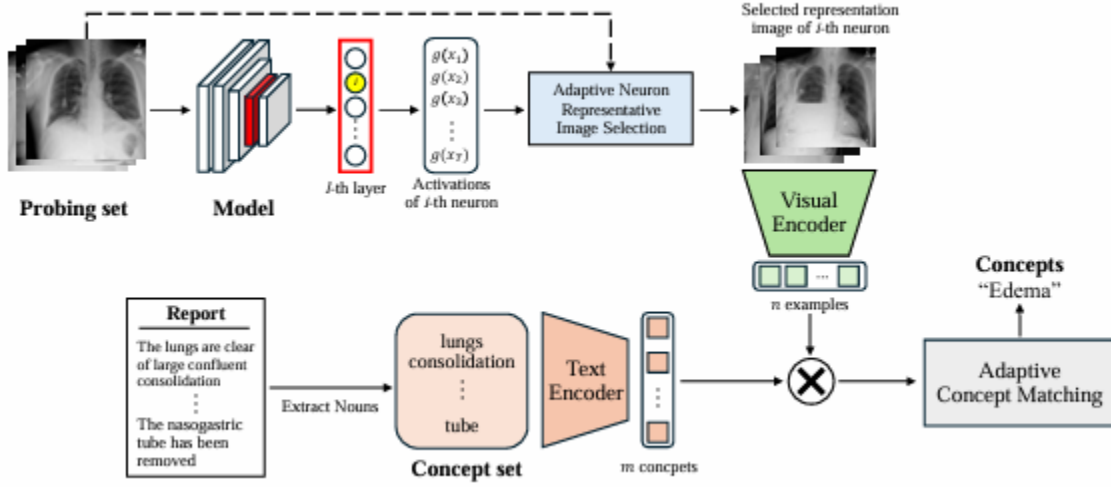


Figure 34.

In the MAMMI framework, chest X-ray images are passed through a target convolutional neural network, DenseNet121, from which neuron activations are extracted. The system then selects representative images for each neuron using an adaptive thresholding mechanism based on the neuron’s activation range, accounting for the class imbalance prevalent in medical datasets. Simultaneously, medical concepts are constructed by extracting domain-relevant nouns from the associated radiology reports. Both image features (from the neurons) and text features (from the concepts) are projected into a shared embedding space using the MedCLIP encoder. Finally, adaptive cosine similarity is computed to associate neurons with the most representative medical concepts. The adaptive threshold (τ) and cosine similarity can be calculated using the equations in Figure Y.

$$\tau_{example}^{(l,i)} = \max(A^{(l,i)}) - (1 - \alpha) \cdot (\max(A^{(l,i)}) - \min(A^{(l,i)})),$$

$$s_j^{(l,i)} = \frac{1}{n} \sum_{o=1}^n \left\{ \cos(v_o^{(l,i)}, t_j) - \cos(v_o^{(l,i)}, t_{tem}) \right\},$$

Figure 36.

Fine-Tuning

The model was cloned from GitHub and used as is. However, we used DenseNet121 as the backbone of the model. This model served as the target for the interpretability method proposed in MAMMI. Rather than modifying the architecture or incorporating additional modules during training, the interpretation was performed independently after training was completed by examining neuron activations. This separation allows the MAMMI method to be applied flexibly to any model of similar architecture without requiring retraining or architectural changes.

Visualisation

After training, the model achieved an AUC score of 0.816, and the mask-free interpretability technique can be seen in Figures A and B

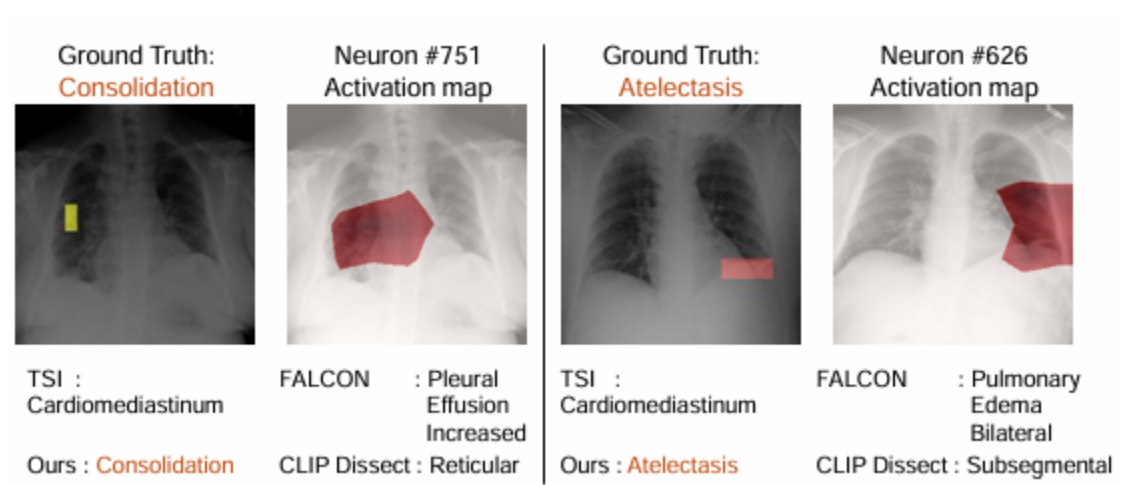


Figure 37.

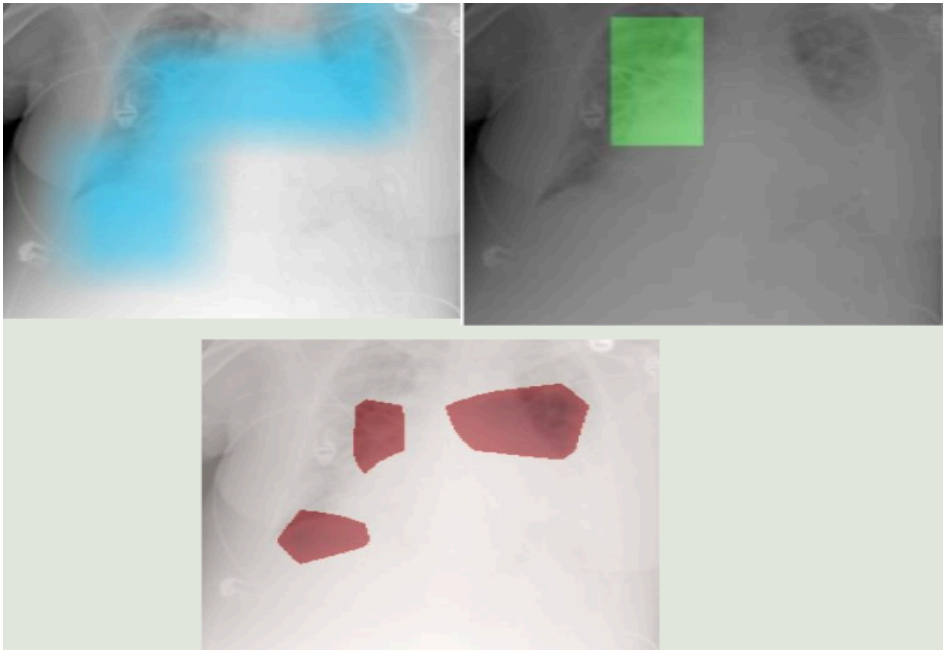


Figure 38. Edema

IV. RESULTS

We evaluated our models with **Area Under The Curve (AUC)** as the primary metric.

Models	Original Paper's AUC	Implementation AUC
KSVM	accuracy = 0.92	accuracy = 0.4
Fine-Tuned Densenet121	0.877	0.75
MobileLungNetV2	0.923	0.71
Fine-Tuned MobileNetV2	0.810	0.72
Dense Convolution Network	0.7821	0.496
Custom CNN	accuracy = 0.95	accuracy = 0.42
Residual	0.8136	0.6179
SwinChex	0.81	0.795
AutoEncoder CNN	0.843	0.9987
MLRFNet	0.853	0.519
CRAL(Attention Mechanism)	0.812	0.6070
Medical Explainability Techniques	-	-
MAMMI	0.881	0.816

V. Discussion

This discussion section provides a comprehensive reflection on a study evaluating machine learning (ML) and deep learning (DL) models for classifying thoracic diseases using the NIH ChestX-Ray14 dataset. Below, I analyze the key takeaways, challenges, explainability techniques, comparisons with prior work, and practical implications, demonstrating critical thinking as requested.

Key Takeaways from Results

The study tested models like Kernel SVM, DenseNet121, MobileNetV2, SwinCheX, MLRFNet, and MAMMI, achieving AUC scores between 0.519 and 0.9987. Notably, AE-CNN and DenseNet121 excelled even with limited data, suggesting that certain architectures are particularly robust for medical imaging tasks. However, performance varied significantly across diseases, with rare conditions like Hernia showing suboptimal results due to insufficient data representation. This highlights a critical insight: model success is heavily tied to data availability and balance, a common issue in medical AI.

Challenges Encountered

Several challenges emerged during the study:

Label Imbalance: Rare diseases were underrepresented, leading to poor model performance. This reflects a broader difficulty in training generalizable models for less common conditions.

Multi-Label Complexity: Interactions between multiple disease labels complicated predictions, as seen in DenseNet121's struggles despite its high AUC for single-label tasks like pneumonia.

Data Quality Issues: Age anomalies and potential labelling errors in multi-label images required extensive preprocessing. This underscores the need for meticulous data curation in medical datasets, as flawed inputs can skew even the best models.

These challenges suggest that while advanced models can achieve high accuracy, their reliability hinges on addressing foundational data issues.

Effectiveness of Explainability Techniques

The study employed XAI techniques like Grad-CAM and LIME to provide visual explanations. These methods proved effective in aligning model focus with clinical expectations, for instance, highlighting lung fields for pneumonia detection. However, their limitations became apparent with rare diseases, where model attention was inconsistent, likely due to imbalanced data. This indicates that while XAI enhances transparency, its utility is constrained by the underlying model's training quality. Improving XAI for edge cases remains an open challenge.

VI. CONCLUSION

This research explored the application of Explainable Artificial Intelligence (XAI) techniques to Machine Learning (ML) and Deep Learning (DL) models for classifying thoracic diseases using chest X-ray images from the NIH Chestx-Ray14 dataset. Various models were assessed, yielding AUC scores between 0.519 and 0.9987. While some models demonstrated exceptional accuracy, challenges such as label imbalance and multi-label complexity were evident, particularly for rare diseases. XAI techniques like Grad-CAM, LIME, and saliency maps facilitated model decisions, improving transparency and trust in clinical settings.

Incorporating XAI is essential for AI-driven diagnostics, as it addresses the interpretability gap that often undermines confidence in black-box models. By providing visual and conceptual explanations, XAI enhances accountability and allows clinicians to verify AI outputs, critical in high-stakes medical applications. This study highlights the significance of explainability, not only in improving diagnostic reliability but also in advancing the adoption of AI technologies in healthcare research and industry.

Looking forward, future efforts should prioritise addressing class imbalance through advanced sampling methods and developing more robust multi-label classification frameworks. Clinical trials to validate model performance in real-world settings are recommended to ensure practical utility. Additionally, investigating hybrid architectures, such as combining Convolutional Neural Networks (CNNs) with Transformers, could further elevate both accuracy and interpretability, paving the way for more effective and trustworthy AI solutions in medical imaging.

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